



# Marin County Large Scale Stormwater Trash Capture Feasibility Report



June 2022

This submittal has been prepared under the direct supervision of the undersigned, who hereby certifies that she is a Registered Professional Engineer in the State of California

**Schaaf & Wheeler**  
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## **Attachments**

Attachment 1: Site Summary Sheets

Attachment 2: RWQCB Approved Device List

Attachment 3: Detailed Costs and Calculations (provided electronically)

Attachment 4: Permitting Matrix

Attachment 5: GIS Shapefile for Small Device Installation (provided electronically)



## 1. Introduction

With the adoption of the Trash Amendments by the State Water Resources Control Board (State Water Board) in April 2015, the implementation of stormwater trash control programs has become a high priority compliance issue for owners/operators of Municipal Separate Storm Sewer Systems (MS4s) throughout the State of California, including Marin County (County). Trash control programs are designed to demonstrate a reduction in trash and litter discharged from MS4s and to protect the water quality and beneficial uses of local receiving water bodies (e.g., creeks, rivers, lakes, San Francisco Bay and the Pacific Ocean) from the impacts of trash. The Trash Amendments require that control programs be implemented to comply with trash discharge prohibitions and water quality objectives, which state that trash from MS4s cannot be discharged in levels that impact beneficial uses of receiving waters.

The jurisdictions in Marin County have selected the Track 1 compliance strategy, which will entail the installation and maintenance of certified full trash capture (FTC) systems to address trash from Priority Land Uses (PLUs). Jurisdictions in Marin County are supported by the Marin Countywide Stormwater Pollution Prevention Program (MCSTOPPP). Previous work was conducted by MCSTOPPP staff to identify PLU areas that have the potential to generate trash. This work also identified 15 potential locations for large scale FTC devices.

This Trash Capture Feasibility study further investigated the fifteen potential locations for feasibility. These locations were prioritized because they capture runoff from California Department of Transportation (Caltrans) right-of-way. Caltrans is under a consent decree with the Regional Water Quality Control Board (RWQCB) and is seeking opportunities to fund local trash capture projects if they treat Caltrans runoff. Cost estimates for the installation and maintenance of these devices were also developed along with site alternatives and permitting implications. Field visits were conducted at each site to develop suitable locations and devices. Anticipated costs of recommended devices and alternatives were also identified. These large scale device locations are compared to the costs and maintenance of small scale devices to capture the equivalent PLUs draining to the large scale device. It should be noted that small scale devices do not capture runoff from Caltrans right-of-way and are thus not able to obtain funding support from Caltrans.

This technical memorandum summarizes the results of the tasks completed via this study and identifies preferred and alternative sites, device options, and installation and maintenance costs so that MCSTOPPP has a tool that will serve as an implementation and funding plan to meet requirements for trash reduction in stormwater. The main deliverable of this study is the one-page site summaries contained in Attachment 1 that summarize each site. These stand-alone summary documents can be used to start coordinating with Caltrans for potential funding. One of the initial sites was broken into two alternative locations so there is a total of sixteen sites reported on in this study.

All trash reduction benefits and costs are based on the best information available at the time the Report was developed. Schaaf & Wheeler worked closely with subconsultants EOA, Inc. and





NCE to develop this Report. The Report presents the best alternatives, associated costs, and alternatives for MCSTOPPP to assist the Marin County jurisdictions in achieving the trash reduction goals outlined in the Trash Amendments.

## 2. Full Trash Capture Devices

### 2.1 Capital Improvement Project Alternatives

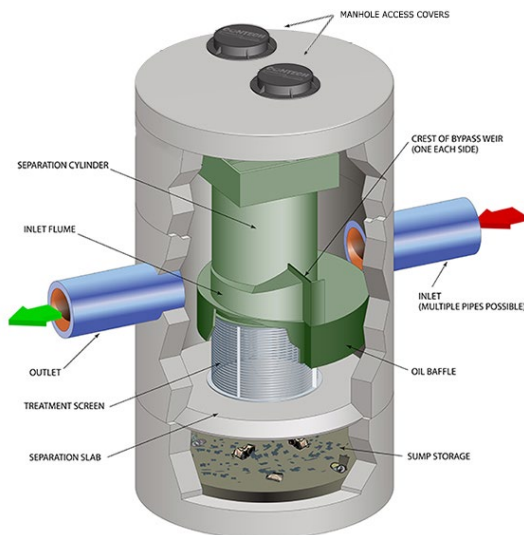
There are a variety of devices available to meet the full capture trash reduction goal, including:

- Catch basin inlet filters/screens;
- Netting, screens, or hydrodynamic structures on a storm drain trunk line; and
- End-of-pipe netting, screens, or hydrodynamic structures.

When selecting a device, it is important to consider the drainage area, hydraulic losses across the system when full, permitting requirements in open channels, device maintenance and access, property rights, and utility clearance. All of these factors should be weighed against the device, installation, and maintenance costs to ensure maximum capture volume per dollar spent. The current list of approved full trash capture devices is contained in Attachment 2.

#### 2.1.1 Large Scale Devices

This study focuses on large scale devices as they have the capability of capturing runoff from Caltrans right-of-way and thus enabling possible funding sources. Large-scale devices provide a centralized location for construction and maintenance. The downside of large devices is that they capture flow from low trash generating areas resulting in the need to have a larger device to treat the additional runoff from areas with little trash reduction benefit.



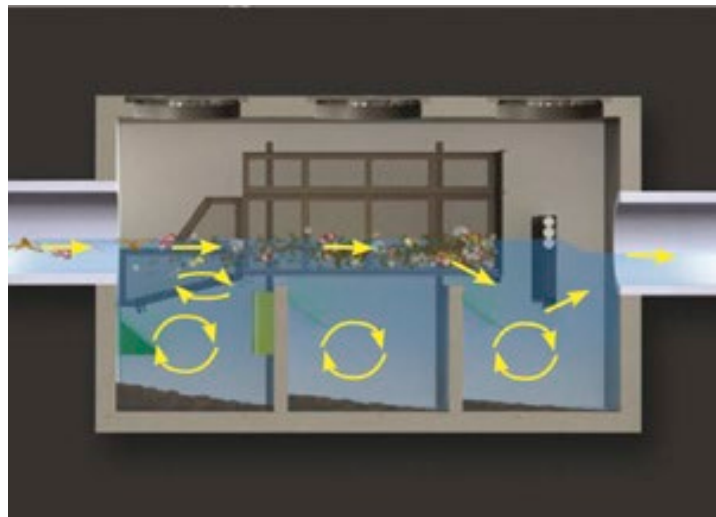
**Hydrodynamic separators (HDS)** are widely used in stormwater treatment. They are flow-through structures with a settling or separation unit to remove sediments, floatables, and other pollutants. They come in a wide range and some are as small as a standard manhole structure. These inline systems can be cleaned using a vacuum truck to pump out the trash, sediment, and water that has collected in the bottom. Engineering and installation costs can be quite high, but the devices are long-lasting and can capture trash from a considerable area.

**Netting devices** can be placed on discharge pipes that will collect trash. Netting systems have one or more mesh bags and a metal frame guide system to support the nets. When full, the nets can be removed and replaced with new nets. End-of-pipe systems can be installed at the actual discharge point to receiving water, or just a collection point. Because these nets are installed above ground, they must be inspected at regular intervals to find and repair any damage. The netting devices may also require permitting for installation and maintenance if they are located in the channel.



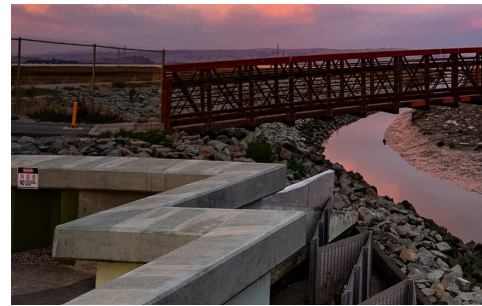
**Baffle boxes** are concrete or fiberglass structures containing a series of sediment settling chambers separated by baffles. The primary function of baffle boxes is to remove sediment, suspended particles, and associated pollutants from stormwater. Trash is removed through the metal cage with 5mm mesh located at the flow line of the storm drain system. Baffle boxes can be located either in-line or at the end of storm pipes.

Cleaning of baffle boxes can be accessed through a manhole or hatch above ground with a vacuum truck, with no confined spaced entry required for servicing. Screened systems are hinged for easy access to collected sediments in the baffle chambers below.



### 2.1.2 Louvered Panels

Louvered panels can be used in small scale and large scale applications. The most common louvered panel is the Rosco Moss Storm Flo® pictured below on the left. Louvered panels can also be used in sites that don't have a traditional catch basin and are sold by several manufacturers in just flat panel sheets of 5mm openings. The screen can be used to span across a channel of irregular shape or at the outfall of a pump station as shown below on the right in a zig zag formation to increase opening.



### 2.1.3 Small Scale Devices

Small scale devices include structures such as connector pipe screens and hanging baskets installed at storm drain inlets. For each large-scale device, we have prepared cost estimates for the equivalent trash capture using small scale devices. However, the small scale devices do not capture the Caltrans runoff as small-scale devices cannot be located on the major trunklines.

The connector pipe screen is a metal screen assembly installed inside a catch basin, in front of the outlet pipe, preventing debris from entering the storm drain system. They are designed to be permanently mounted to the catch basin or have a quick disconnect feature to facilitate pipe jetting or to prevent flooding should the basin drainage system become clogged. These units are designed to retain all trash and solids larger than 5 mm inside the catch basin, and once in place, it also retains large volumes of sediment. Therefore, the trash capture and storage capacity of this product are dependent upon the size of the catch basin where it is installed.



The hanging basket is similar to the connector pipe screen in that it is a separate unit that is installed into the storm drain system. Typically, installed in a catch basin, the hanging basket is an insert that keeps trash and large debris from entering the stormwater system. A frame is inserted into the top of the catch basin and is paired with a basket made with 5mm opening perforations. The large screen openings allow ample flow through and will not clog from sand and sediment loading off streets and parking lots. Inlet screens are desirable because they can be installed relatively quickly and at a relatively low cost.



## 3. Cost Estimates and Maintenance Considerations

### 3.1 Unit Costs

The following assumptions were made in determining construction costs for each proposed device:

- Installation costs for large-scale HDS and baffle devices are assumed to be 4 times the cost of the device;
- Installation of netting large-scale devices are assumed to be 3 times the cost of the device;
- Installation of louvered panels was calculated on a case-by-case basis as they are custom installations;
- Hydrodynamic separators and baffle boxes capital cost is estimated at \$3000 per treatment flow rate (cfs) based on recent bids;
- Netting and large screen device capital costs are estimated at \$2500 per treatment flow rate (cfs) based on recent bids;
- Small scale devices (i.e. inlet screens and hanging baskets) were estimated at \$2,500 per device, based on recent quotes;
- The devices are located in areas which do not require major utility relocation, property or easement acquisition or environmental permitting;
- Large scale devices were assumed to have a lifetime of 50-years;
- Small scale devices were assumed to have a lifetime of 25-years;
- Total Base Cost is presented in the 2022 construction year dollars and includes installation, construction needed to update drainage infrastructure to support trash capture device, and the device. It does not include maintenance or device replacement;
- 20% construction contingency added to total costs
- 15% design added
- Permitting costs were estimated for each specific site location.

Table 4-1 lists the unit costs used to develop each of the site's installation cost. Attachment 3 contains details of each site quantities and total cost.



**Table 3-1: Unit Cost Assumptions**

| Item                       | Cost per Unit         | Unit     |
|----------------------------|-----------------------|----------|
| HDS or Baffle              | \$3,000               | per cfs  |
| Netting Device             | \$2,500               | per cfs  |
| Louvered Panel             | \$500                 | SF       |
| Connector Pipe Screen      | \$2,500               | each     |
| Permitting for Underground | \$27,000              | lump sum |
| Permitting for in channel  | \$105,000 - \$120,000 | lump sum |

### 3.2 Maintenance Considerations

Cleaning and maintenance costs are not included in the installation costs. They are provided in this section for reference only for the MCSTOPPP consideration. Maintenance of both small-and large-scale devices can be done by municipal staff or hired out to a cleaning contractor. In the Bay Area, there is a mix of jurisdictions that perform maintenance in-house and those that contract out the maintenance.

For cleaning, replacement, and maintenance, the following considerations should be taken:

- Large scale HDS or baffle box devices are cleaned approximately three (3) times a year, requiring 8 hours of a team of two staff members at \$100/hour, a vacuum truck at \$400/hour, or \$23,400/device annually;
- Netting devices are cleaned approximately three (3) times a year, requiring 8 hours of a team of two staff members at \$100/hour, a crane truck at \$5,000/half day, replacement net at \$1,500, or \$27,300/device annually;
- Louvered panels at the Rossi pump station are cleaned approximately three (3) times a year, requiring 4 hours of a team of two staff members at \$100/hour, a vacuum truck at \$400/hour, or \$7,200/device annually;
- Inlet screens are cleaned approximately three (3) times a year, requiring 4 hours of a team of two maintenance workers at \$100/hour- or \$7,200/device annually;





## 4. Permit Considerations

### 4.1 Overview

Subconsultant NCE assisted with this Study in preparing permitting implications for each site location. Each site location was analyzed against the CEQA Appendix G Environmental checklist topics to determine if the site location would have a potential for a significant effect on the environment. A site visit and background research were conducted for each location to understand the significance of the alternatives on the environment and to determine what permits could be needed from regulatory agencies. Attachment 4 details each site in detail.

### 4.2 Findings

No special status species were identified within the proposed project site locations. However, the potential exists that special status species, including nesting avian species (all sites), could be present at any of the locations if construction activities occur, including during the avian nesting season. Five of the potential sites (LS-028, NV-108, SR-111, SR-129, and ST-003) were identified to have potential habitat that could support special status species in addition to nesting birds.

Aquatic resources were identified within five of the potential sites (LS-028, NV-108, SR-111, SR-129, and ST-003). Construction activities at these five sites would result in impacts to aquatic resources and would require Marin County to obtain regulatory permits and clearances from the State Water Resources Control Board (SWQRCB), California Department of Fish and Wildlife (CDFW), San Francisco Bay Conservation and Development Commission (BCDC), and the U.S. Army Corps of Engineers (USACE). Regulatory permitting costs savings can be achieved by combining multiple sites into single projects or bundles and preparing/submitting a single permitting package rather than multiple permitting packages.

A Record Search from the Northwest Information Center was conducted to determine the potential presence of cultural resources within the vicinity of each location and no resources were identified. Additionally, a Sacred Lands file search from the Native American Heritage Commission (NAHC) was conducted. The NAHC database search response was received on March 11, 2022 and indicated that there are Native American Resources in the vicinity of the sites. The response, however, did not indicate where the site(s) are located. Further correspondence with the Graton Rancheria will be required to determine if any sites are near a resource.

The Feasibility Analysis Matrix (Attachment 4) describes the findings of the site visits and research conducted for each site for the following:

- Biological Resources
- Cultural Resources
- Geology and Soils
- Hydrology and Water Quality
- Transportation



- Tribal Cultural Resources
- Utilities

The Feasibility Analysis Matrix also identifies potential future costs to conduct additional technical studies, prepare technical reports, and what regulatory permits may be required to construct the trash collectors at the sites, and the costs to prepare and secure the said permits.

### **4.3 California Environmental Quality Act (CEQA) Compliance**

Five of the proposed sites are expected to require the preparation of a full mitigated negative declaration (MND) level document to comply with the requirements of CEQA. The remaining 11 sites are expected to only require a categorical exclusion level CEQA document, which is much less involved than a MND due to less identified environmental impacts. CEQA compliance costs for a categorical exclusion are approximately \$5,000, while a MND would cost approximately \$28,000. Environmental review cost savings can be achieved by combining multiple sites into single projects or bundles and conducting a single review rather than multiple review.





## 5. Recommendations

### 5.1 Coordination with RWQCB

The Trash Amendments were adopted in 2015, but have not yet been incorporated into the Phase II NPDES permit, which was scheduled for reissuance in 2021, but has been delayed due to COVID. Until the Phase II permit is reissued, Marin County permittees have no trash reduction or reporting requirements associated with the Trash Amendments. That said, the Trash Amendments include a deadline of 2030 when trash from all PLUs needs to be controlled via full trash capture systems or measures that are full capture system equivalent (FSCE). Therefore, it is recommended that permittees begin developing a funding plan and utilize the information included in this Study to identify which devices should be installed in the upcoming few years.

Additionally, it is recommended that MCSTOPPP staff begin verbal communications with Regional Water Board staff soon to update staff on the steps that MCSTOPPP has taken to identify and install full trash capture systems. By communicating early with the Regional Water Board staff, the Marin County permittees will be best positioned to adjust its implementation program as needed based on the perspectives of the regulatory agency overseeing the implementation of the NPDES permit. Communications could include sharing the final version of this report and/or providing a presentation of the tasks completed to-date and the planned installation of full trash capture systems. Part of this study did include a meeting with Regional Staff on the possibility of installing end of pipe netting. While there was no formal approval during the meeting, there was not a formal disapproval.

### 5.2 Prioritized Installation

The metrics that MCSTOPPP will need to generate to demonstrate progress towards fully addressing the Trash Amendment goal will be identified in the reissued Phase II NPDES permit. Table 5-1 lists the 15 sites and details the cost implications. Each site has a dollar per PLU acres treated for both small and large. Finally, the sites were ranked from high priority to low priority. A ranking of 1 is a high priority site that could be installed within the next few years. Sites ranked with a 2-3 could be installed in the next 5 years. The sites ranked 4-5 may be better suited for small scale device installation or require further improvements or coordination with the Regional Board before they can proceed forward.

**Table 5-1: Device Dollars per Treatment Percentage Achieved**

| Site ID       | Device Type        | PLU Treated (acres) | Caltrans Treated (acres) | Total Capital (\$) | \$/PLU Large | \$/PLU Small | Priority Ranking |
|---------------|--------------------|---------------------|--------------------------|--------------------|--------------|--------------|------------------|
| <b>MV-037</b> | Louvered Panel     | 10.74               | -                        | 60,000             | 2,200        | 12,800       | 1                |
| <b>TB-010</b> | Large Scale Device | 17.96               | 5.03                     | 250,000            | 10,800       | 33,400       | 1                |
| <b>TV-001</b> | Large Scale Device | 17.18               | 3.73                     | 320,000            | 14,800       | 20,100       | 1                |



| Site ID       | Device Type        | PLU Treated (acres) | Caltrans Treated (acres) | Total Capital (\$) | \$/PLU Large | \$/PLU Small | Priority Ranking |
|---------------|--------------------|---------------------|--------------------------|--------------------|--------------|--------------|------------------|
| <b>LS-014</b> | Large Scale Device | 15.51               | 13.08                    | 380,000            | 19,500       | 24,800       | 1                |
| <b>SR-143</b> | Louvered Panel     | 35.19               | 15.87                    | 350,000            | 8,000        | 6,800        | 2                |
| <b>CM-016</b> | Large Scale Device | 42.53               | 29.96                    | 1,650,000          | 28,900       | 15,500       | 2                |
| <b>LS-010</b> | Large Scale Device | 36.47               | -                        | 620,000            | 14,200       | 8,300        | 3                |
| <b>NV-029</b> | Large Scale Device | 4.35                | 11.57                    | 220,000            | 38,900       | 6,400        | 3                |
| <b>SS-003</b> | Inlet Screens      | 42.11               | 32.50                    | 30,000             | -            | 5,100        | 4                |
| <b>ST-003</b> | Trash Capture Net  | 36.94               | 29.51                    | 830,000            | 16,600       | 13,200       | 4                |
| <b>SS-002</b> | Large Scale Device | 26.18               | 35.71                    | 1,010,000          | 32,800       | 6,200        | 4                |
| <b>SR-111</b> | Large Scale Device | 7.93                | 8.10                     | 180,000            | 16,900       | 2,000        | 5                |
| <b>LS-028</b> | Trash Capture Net  | 41.17               | 5.43                     | 520,000            | 8,500        | 2,900        | 5                |
| <b>SR-129</b> | Trash Capture Net  | 18.06               | 20.09                    | 420,000            | 15,400       | 4,300        | 5                |
| <b>NV-116</b> | Louvered Panel     | 80.25               | 23.48                    | 1,590,000          | 15,900       | 8,900        | 5                |
| <b>SR-108</b> | Large Scale Device | 54.63               | -                        | 1,960,000          | 30,800       | 11,300       | 5                |

### 5.3 Next Steps

Based on the information provide in this report, we recommend that MCSTOPPP:

- Identify funding for the detailed design and installations of the early installations and then identify how many installs can be achieved within the dedicated/available funding.
- Coordinat with Caltrans for funding support on large-scale devices that address Caltrans right-of-way (ROW).
- Consider installing small-scale devices in watersheds where the large scale device was ranked 4 or 5 and does not appear feasible.

The MCSTOPP should also begin setting aside funding for maintenance and operation needs and begin conducting staff training for device maintenance and operation. Jurisdictions without a vactor truck might need to obtaining one, since this type of equipment will be needed to maintain large-scale (and possibly small-scale) devices. If the permittees elects to install trash



capture netting devices, a crane will be needed as well to maintain these devices. There are also private contractors who can be hired to do the maintenance in lieu of using MCSTOPPP staff and equipment.

# **ATTACHMENT 1**

Site Summary Sheets



CONCEPTUAL DESIGN - POTENTIAL FULL TRASH SYSTEM PROJECT



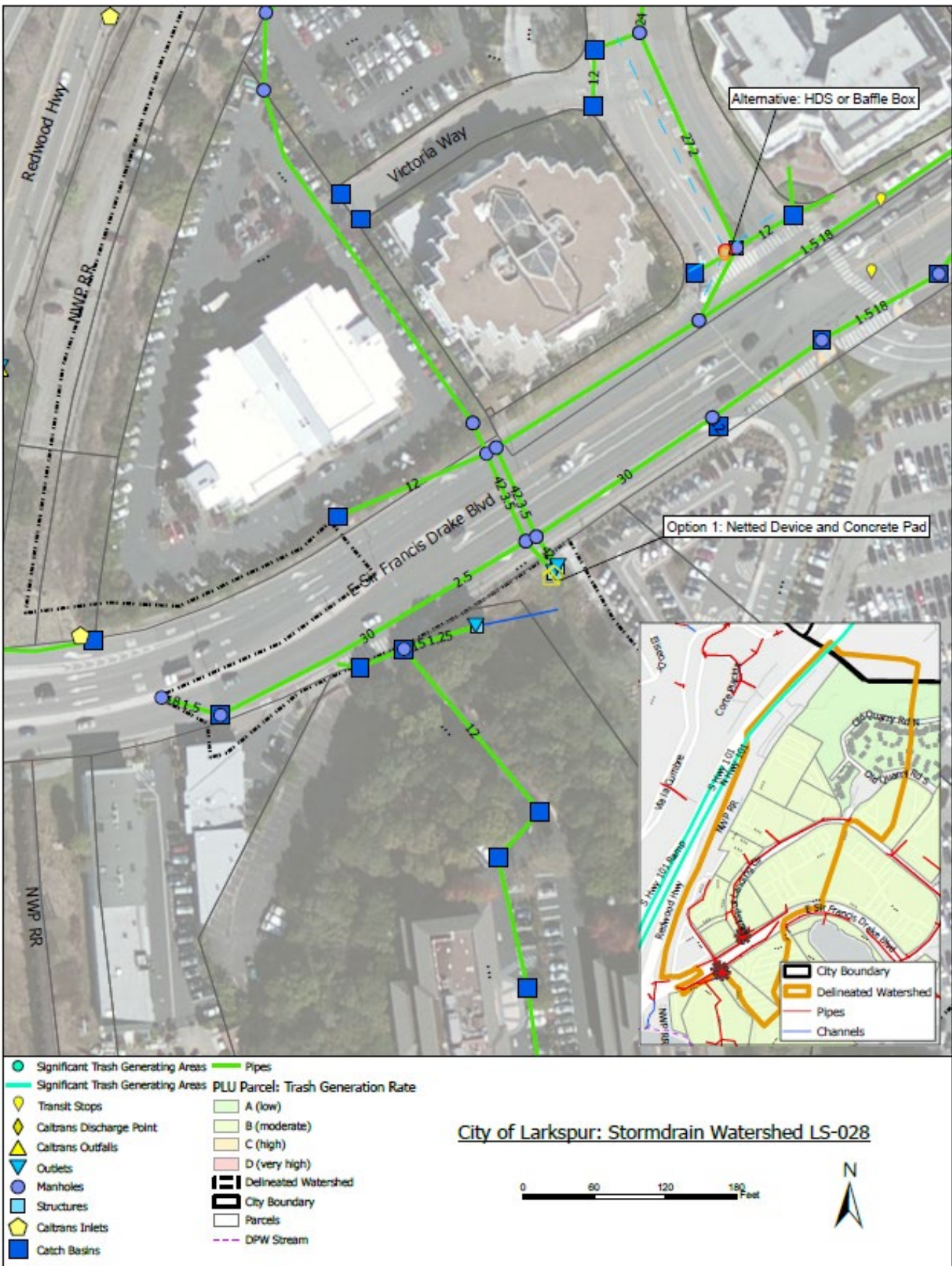
|                                                                                                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Project ID:</b> CM-016-Alt1                                                                                                              |
| <b>Project Name:</b> Town Center                                                                                                            |
| <b>Project Location:</b> Town of Corte Madera ( <a href="#">37.9271, -122.5114</a> )<br>(Project drains Caltrans right-of-way area as well) |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Site/Device Information</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Proposed Device Type:</b> Hydrodynamic Separator (HDS) or Baffle Box and Connector Pipe Screen (CPS) at North End; HDS or Baffle Box at South End (See page 2 for Alternatives)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Project Description:</b> This project would have multiple devices installed to address all Priority Land Use (PLU) areas in the stormwater catchment. The proposed alternative (Alt1) would entail the construction of an underground hydrodynamic separator (HDS) or baffle box unit and catch basin connector pipe screen (CPS) north of the Marina Detention Basin (North End), and the construction of an underground HDS or baffle box unit adjacent to the Marina Detention Basin (South End). The HDS or baffle box unit on the North End would be installed within an existing 48-inch pipe on San Clemente Drive. The CPS unit would be installed in a storm drain catch basin that connects to an 18-inch pipe and is located on Tamalpais Drive west of the HDS or baffle box unit. The second HDS or baffle box unit would be constructed on San Clemente Drive prior the pipe outfall into the detention basin. |
| <b>Site Specific Considerations:</b> <ul style="list-style-type: none"><li>Near tidal water body (North End)</li><li>Existing storm drain infrastructure appears to be &gt;50 years old</li><li>Location of HDS/Baffle Box at North End would impede on paved bike path</li><li>Both HDS/Baffle Boxes would be located on a busy street segment</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>O&amp;M Summary:</b> Requires a vacuum-assisted truck and traffic control/lane closure                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Comments/Notes:</b> Conversations with maintenance staff indicate the existing pipe infrastructure could be in poor condition and may require rehabilitation prior to the install of an HDS or baffle box unit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |



| Drainage Area and Trash Reduction Information                                                                                                           |                           |     |      |           |              |            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-----|------|-----------|--------------|------------|
| <b>Total Drainage Area:</b> 154 acres <ul style="list-style-type: none"><li>Corte Madera – 124 acres</li><li>Caltrans Right-of-Way – 30 acres</li></ul> |                           |     |      |           |              |            |
| <b>Priority Land Use (PLU) Areas (acres):</b> 43 acres                                                                                                  |                           |     |      |           |              |            |
| Public Agency                                                                                                                                           | Trash Generation Category |     |      |           | Non-PLU Area | Total Area |
|                                                                                                                                                         | Low                       | Mod | High | Very High |              |            |
| Corte Madera                                                                                                                                            | 4                         | 35  | 4    | 0         | 81           | 124        |
| Caltrans                                                                                                                                                | --                        | --  | --   | --        | --           | 30         |
| <b>1-yr/1-hr Peak Flow (modeled):</b> 59 cfs                                                                                                            |                           |     |      |           |              |            |

| Preliminary Cost Estimate                 |                    |
|-------------------------------------------|--------------------|
| Design/Permitting                         | \$212,000          |
| Construction                              | \$1,050,000        |
| Construction Contingency                  | \$211,000          |
| <b>Total Capital</b>                      | <b>\$1,650,000</b> |
| <b>Annual Operation &amp; Maintenance</b> | <b>\$23,400/yr</b> |
| <b>Construction Cost per Acre of PLU</b>  | <b>\$29,000/ac</b> |







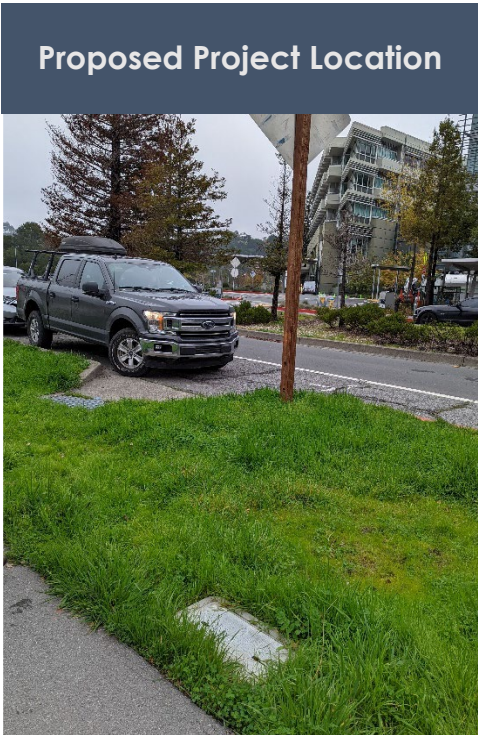
| Alternative Design Options |                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                     |                                                                               |                                                                                                                                                                                                                                                                                                        |                                   |                |                                                                                                                                                                                                                                                                                                              |
|----------------------------|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alternative Site ID        | Proposed Device Type                             | Project Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Site Specific Considerations                                                                                                                                        | O&M Summary                                                                   | Comments/Notes                                                                                                                                                                                                                                                                                         | Preliminary Cost Estimate         |                | Photos                                                                                                                                                                                                                                                                                                       |
| CM-016-Alt2                | Netting Systems at both the South and North Ends | This project would have multiple devices installed to address all Priority Land Use (PLU) areas in the stormwater catchment. This proposed alternative would entail the construction two netting systems. One in the channel on the North End and one in the detention basin at the South End. Both would require the construction of a concrete pad and concrete or metal weir at the outfalls into the channel/basin. Netting devices would be attached to the constructed weir. | <ul style="list-style-type: none"><li>Near tidal water body (North End)</li><li>Dewatering of the channel and basin would be required during construction</li></ul> | May require drawdown if standing water is present when maintenance is planned | Project will require coordination with the SF Bay Regional Water Board (Water Board), which oversees water quality certification programs for creeks/channels in the Bay Area. The project is likely dependent on permission from the Water Board to install a netting system in the northern channel. | Design/Permitting                 | \$230,000      | <div>Proposed Project Location</div>  <div>Northern Location</div> <div>Proposed Project Location</div>  <div>Southern Location</div> |
|                            |                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                     |                                                                               |                                                                                                                                                                                                                                                                                                        | Construction                      | \$590,000      |                                                                                                                                                                                                                                                                                                              |
|                            |                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                     |                                                                               |                                                                                                                                                                                                                                                                                                        | Construction Contingency          | \$117,000      |                                                                                                                                                                                                                                                                                                              |
|                            |                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                     |                                                                               |                                                                                                                                                                                                                                                                                                        | Total Capital                     | \$960,000      |                                                                                                                                                                                                                                                                                                              |
|                            |                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                     |                                                                               |                                                                                                                                                                                                                                                                                                        | Annual Operation & Maintenance    | \$27,300/yr    |                                                                                                                                                                                                                                                                                                              |
|                            |                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                     |                                                                               |                                                                                                                                                                                                                                                                                                        | Construction Cost per Acre of PLU | \$22,500/ac    |                                                                                                                                                                                                                                                                                                              |
| CM-016-Alt3                | Small Devices                                    | This proposed alternative would entail the construction of approximately 220 inlet screens throughout the drainage area of this project.                                                                                                                                                                                                                                                                                                                                           | N/A                                                                                                                                                                 | Requires maintenance at many different locations.                             | Construction Contingency cost included in the overall Construction cost.                                                                                                                                                                                                                               | Design/Permitting                 | \$110,000      | N/A                                                                                                                                                                                                                                                                                                          |
|                            |                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                     |                                                                               |                                                                                                                                                                                                                                                                                                        | Construction                      | \$550,000      |                                                                                                                                                                                                                                                                                                              |
|                            |                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                     |                                                                               |                                                                                                                                                                                                                                                                                                        | Total Capital                     | \$660,000      |                                                                                                                                                                                                                                                                                                              |
|                            |                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                     |                                                                               |                                                                                                                                                                                                                                                                                                        | Annual Operation & Maintenance    | \$1,580,000/yr |                                                                                                                                                                                                                                                                                                              |
|                            |                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                     |                                                                               |                                                                                                                                                                                                                                                                                                        | Construction Cost per Acre of PLU | \$15,500/ac    |                                                                                                                                                                                                                                                                                                              |



CONCEPTUAL DESIGN - POTENTIAL FULL TRASH SYSTEM PROJECT



|                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Project ID: LS-010-Alt1                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Project Name: Bon Air                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Project Location: Marin County ( <a href="#">37.9468, -122.5376</a> )                                                                                                                                                                                                                                                                                                                                                                                          |
| Site/Device Information                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Proposed Device Type: Hydrodynamic Separator (HDS) or Baffle Box and Green Infrastructure (See page 2 for Alternatives)                                                                                                                                                                                                                                                                                                                                        |
| Project Description: The proposed alternative (Alt1) would entail the construction of an underground hydrodynamic separator (HDS) or baffle box unit within a catch basin located on the side of Bon Air Road. The HDS or baffle box unit would be installed approximately 10-feet underground within an existing 72-inch pipe on Bon Air Road.                                                                                                                |
| Site Specific Considerations: <ul style="list-style-type: none"><li>Near tidal water body</li><li>Existing storm drain infrastructure appears to be &gt;50 years old</li><li>Location of HDS/Baffle Box is on the side of a road next to street parking area and paved bike path</li><li>Street light box located nearby</li></ul>                                                                                                                             |
| O&M Summary: Requires a vacuum-assisted truck and traffic control/lane closure                                                                                                                                                                                                                                                                                                                                                                                 |
| Comments/Notes: Conversations with maintenance staff indicate the existing pipe infrastructure could be in poor condition and may require rehabilitation prior to the install of an HDS or baffle box unit. Project site does not include any Caltrans drainage area and might be better suited for Caltrans funding for green infrastructure to offset Sonoma-Marin narrows project. The project will require coordination with the Roads and Parks division. |



| Drainage Area and Trash Reduction Information                                                                                                 |                           |     |      |           |              |            | Preliminary Cost Estimate         |              |
|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-----|------|-----------|--------------|------------|-----------------------------------|--------------|
| Total Drainage Area: 80 acres <ul style="list-style-type: none"><li>Marin County – 80 acres</li><li>Caltrans Right-of-Way – 0 acres</li></ul> |                           |     |      |           |              |            | Design/Permitting                 | \$105,000    |
| Priority Land Use (PLU) Areas (acres): 36 acres                                                                                               |                           |     |      |           |              |            | Construction                      | \$356,000    |
| Public Agency                                                                                                                                 | Trash Generation Category |     |      |           | Non-PLU Area | Total Area | Construction Contingency          | \$71,200     |
|                                                                                                                                               | Low                       | Mod | High | Very High |              |            | Total Capital                     | \$624,000    |
| Marin County                                                                                                                                  | 1                         | 17  | 16   | 2         | 44           | 36         | Annual Operation & Maintenance    | \$23,000/yr  |
| Caltrans                                                                                                                                      | -                         | -   | -    | -         | -            | 0          | Construction Cost per Acre of PLU | \$14,200/ ac |
| 1-yr/1-hr Peak Flow (modeled): 30 cfs                                                                                                         |                           |     |      |           |              |            |                                   |              |







| Alternative Design Options |                                                            |                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                      |                                                                                           |                                                                                                                                                                                                                                                                                               |                                   |              |                                                                                                                                                           |
|----------------------------|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alternative Site ID        | Proposed Device Type                                       | Project Description                                                                                                                                                                                                                                                                                                   | Site Specific Considerations                                                                                                                                                                                                         | O&M Summary                                                                               | Comments/Notes                                                                                                                                                                                                                                                                                | Preliminary Cost Estimate         |              | Photos                                                                                                                                                    |
| LS-010-Alt2                | Netting System                                             | This proposed alternative would entail the construction one netting system in the channel adjacent to Bon Air Road. The netting system would require the construction of a concrete pad and concrete or metal weir at the outfalls into the channel. Netting devices would be attached to the constructed weir.       | <ul style="list-style-type: none"><li>Near tidal water body (North End)</li><li>Dewatering of the channel and basin would be required during construction</li></ul>                                                                  | May require dewatering/drawdown if standing water is present when maintenance is planned. | Project will require coordination with the SF Bay Regional Water Board (Water Board), which oversees water quality certification programs for creeks/channels in the Bay Area. The project is likely dependent on permission from the Water Board to install a netting system in the channel. | Design/ Permitting                | \$175,500    | <div>Proposed Project Location</div>  <div>Northern Location</div>     |
|                            |                                                            |                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                      |                                                                                           |                                                                                                                                                                                                                                                                                               | Construction                      | \$297,000    |                                                                                                                                                           |
|                            |                                                            |                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                      |                                                                                           |                                                                                                                                                                                                                                                                                               | Construction Contingency          | \$59,000     |                                                                                                                                                           |
|                            |                                                            |                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                      |                                                                                           |                                                                                                                                                                                                                                                                                               | Total Capital                     | \$545,500    |                                                                                                                                                           |
|                            |                                                            |                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                      |                                                                                           |                                                                                                                                                                                                                                                                                               | Annual Operation & Maintenance    | \$27,300/yr  |                                                                                                                                                           |
|                            |                                                            |                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                      |                                                                                           |                                                                                                                                                                                                                                                                                               | Construction Cost per Acre of PLU | \$15,000/ac  |                                                                                                                                                           |
| LS-010-Alt3                | Green Infrastructure with Trash Capture on Overflow Inlets | This proposed alternative would entail the construction of approximately 101 inlet screens throughout the drainage area of this project. Additionally, green infrastructure should be installed in the median on Bon Air Road, estimated cost is only for inlet screens and does not include the green infastructure. | <ul style="list-style-type: none"><li>Site does not qualify for Caltrans trash funding, but could obtain Caltrans funding for green infrastructure as part of the mitigation measures for the Marin Sonoma Narrows Project</li></ul> | Requires maintenance at many different locations.                                         | Construction Contingency cost included in the overall Construction cost. Cost estimate does not include the cost of green infrastructure.                                                                                                                                                     | Design/ Permitting                | \$50,500     | <div>Proposed Project Location</div>  <div>Bon Air Road Median</div> |
|                            |                                                            |                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                      |                                                                                           |                                                                                                                                                                                                                                                                                               | Construction                      | \$253,000    |                                                                                                                                                           |
|                            |                                                            |                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                      |                                                                                           |                                                                                                                                                                                                                                                                                               | Total Capital                     | \$303,000    |                                                                                                                                                           |
|                            |                                                            |                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                      |                                                                                           |                                                                                                                                                                                                                                                                                               | Annual Operation & Maintenance    | \$727,000/yr |                                                                                                                                                           |
|                            |                                                            |                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                      |                                                                                           |                                                                                                                                                                                                                                                                                               | Construction Cost per Acre of PLU | \$8,308/ac   |                                                                                                                                                           |



CONCEPTUAL DESIGN - POTENTIAL FULL TRASH SYSTEM PROJECT



| Project ID: LS-014-Alt1                                                                                                                                                                                                 |                           |     |      |           |              |            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-----|------|-----------|--------------|------------|
| Project Name: Black Kettle Lagoon                                                                                                                                                                                       |                           |     |      |           |              |            |
| Project Location: Town of Corte Madera ( <a href="#">37.9391</a> , <a href="#">-122.5168</a> )<br>(Project drains Caltrans right-of-way area as well)                                                                   |                           |     |      |           |              |            |
| Site/Device Information                                                                                                                                                                                                 |                           |     |      |           |              |            |
| Proposed Device Type: Hydrodynamic Separator (HDS) or Baffle Box (See page 2 for Alternatives)                                                                                                                          |                           |     |      |           |              |            |
| Project Description: Construction of an underground hydrodynamic separator (HDS) or baffle box unit within an existing 3-foot by 4-foot box culvert. The device will be located upstream of the lagoon.                 |                           |     |      |           |              |            |
| Site Specific Considerations: <ul style="list-style-type: none"><li>Located near a lagoon</li></ul>                                                                                                                     |                           |     |      |           |              |            |
| O&M Summary: Requires a vacuum-assisted truck and traffic control/lane closure.                                                                                                                                         |                           |     |      |           |              |            |
| Comments/Notes: It is unclear whether the pipes from the south and west connect to the box culvert. Additional connector pipe screen devices may be required on smaller pipes that don't connect to the larger culvert. |                           |     |      |           |              |            |
| Drainage Area and Trash Reduction Information                                                                                                                                                                           |                           |     |      |           |              |            |
| Total Drainage Area: 37 acres <ul style="list-style-type: none"><li>Corte Madera – 24 acres</li><li>Caltrans Right-of-Way – 13 acres</li></ul>                                                                          |                           |     |      |           |              |            |
| Priority Land Use (PLU) Areas (acres): 16 acres                                                                                                                                                                         |                           |     |      |           |              |            |
| Public Agency                                                                                                                                                                                                           | Trash Generation Category |     |      |           | Non-PLU Area | Total Area |
|                                                                                                                                                                                                                         | Low                       | Mod | High | Very High |              |            |
| Corte Madera                                                                                                                                                                                                            | 0                         | 12  | 3    | 0         | 8            | 24         |
| Caltrans                                                                                                                                                                                                                | --                        | --  | --   | --        | --           | 13         |
| 1-yr/1-hr Peak Flow (modeled): 18 cfs                                                                                                                                                                                   |                           |     |      |           |              |            |

Proposed Project Location

Lagoon Near Project Site

Proposed Project Location

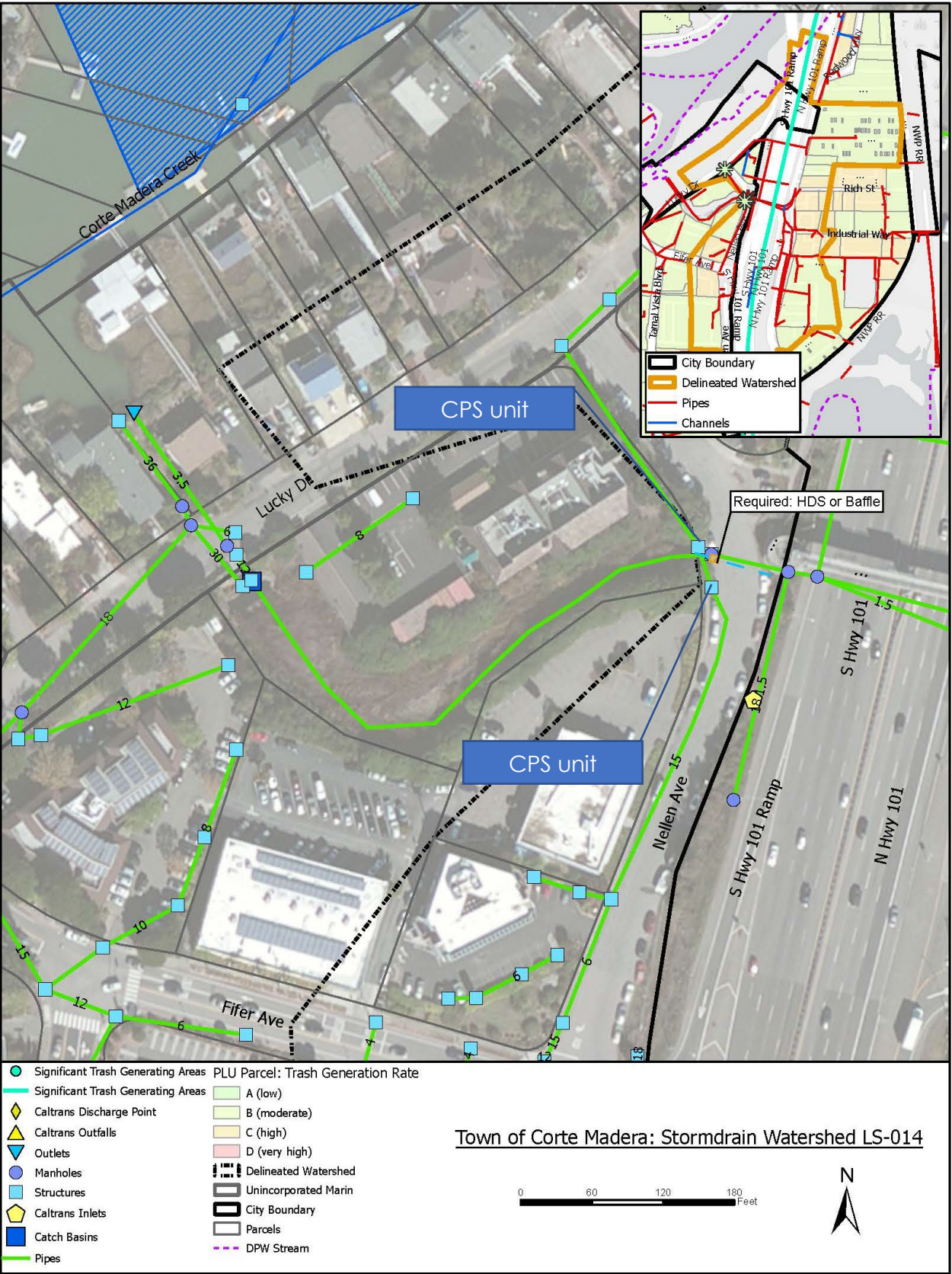


Lagoon Near Project Site



Preliminary Cost Estimate

|                                   |             |
|-----------------------------------|-------------|
| Design/Permitting                 | \$72,000    |
| Construction                      | \$212,000   |
| Construction Contingency          | \$42,400    |
| Total Capital                     | \$375,500   |
| Annual Operation & Maintenance    | \$23,400/yr |
| Construction Cost per Acre of PLU | \$19,500/ac |





CONCEPTUAL DESIGN - POTENTIAL FULL TRASH SYSTEM PROJECT



| Alternative Design Options |                      |                                                                                                                                          |                              |                                                   |                                                                          |                                   |              |        |
|----------------------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|---------------------------------------------------|--------------------------------------------------------------------------|-----------------------------------|--------------|--------|
| Alternative Site ID        | Proposed Device Type | Project Description                                                                                                                      | Site Specific Considerations | O&M Summary                                       | Comments/Notes                                                           | Preliminary Cost Estimate         |              | Photos |
| LS-014-Alt2                | Small Devices        | This proposed alternative would entail the construction of approximately 128 inlet screens throughout the drainage area of this project. | N/A                          | Requires maintenance at many different locations. | Construction Contingency cost included in the overall Construction cost. | Design/ Permitting                | \$64,000     | N/A    |
|                            |                      |                                                                                                                                          |                              |                                                   |                                                                          | Construction                      | \$320,000    |        |
|                            |                      |                                                                                                                                          |                              |                                                   |                                                                          | Total Capital                     | \$384,000    |        |
|                            |                      |                                                                                                                                          |                              |                                                   |                                                                          | Annual Operation & Maintenance    | \$922,000/yr |        |
|                            |                      |                                                                                                                                          |                              |                                                   |                                                                          | Construction Cost per Acre of PLU | \$24,800/ac  |        |



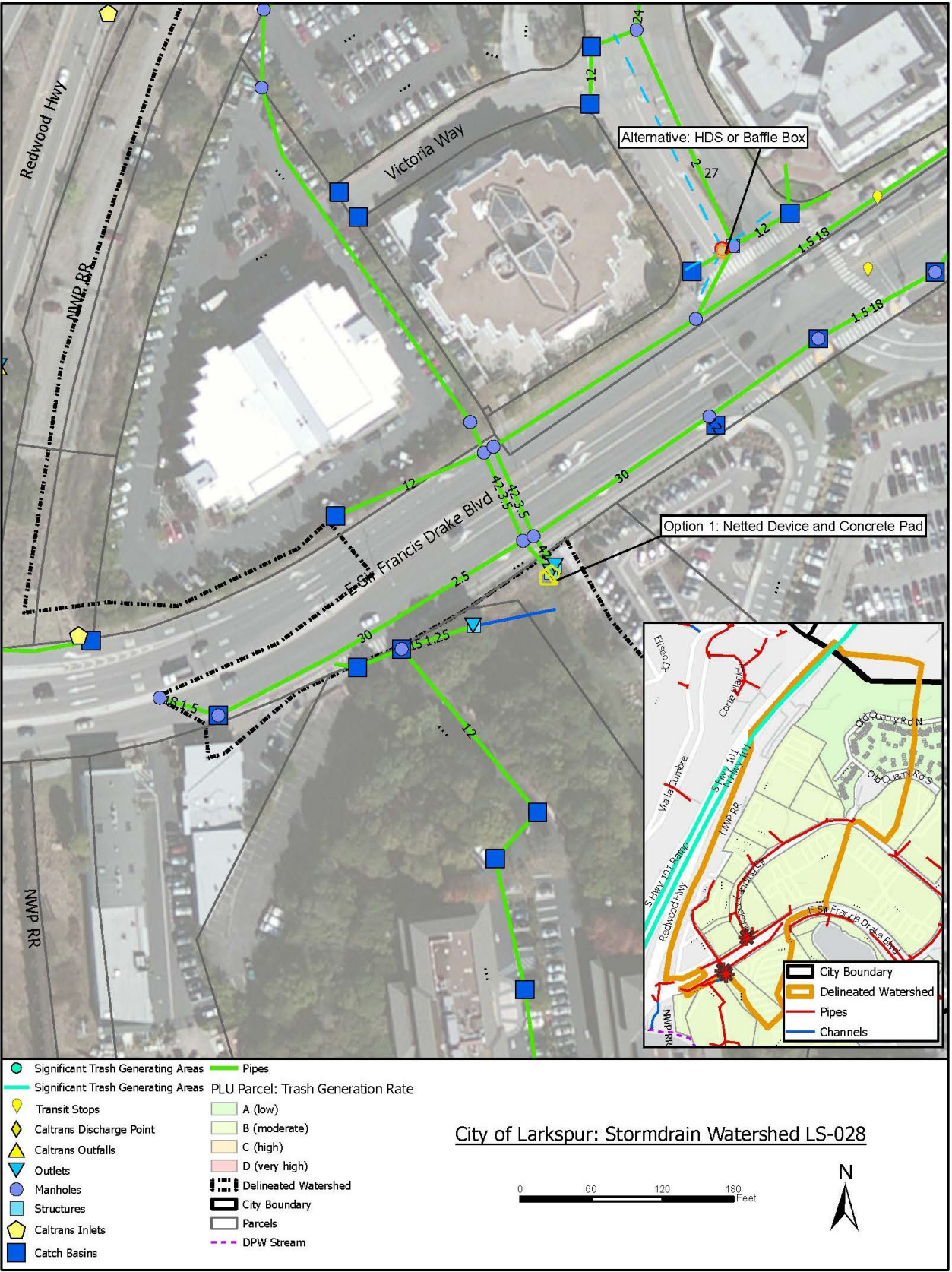
CONCEPTUAL DESIGN - POTENTIAL FULL TRASH SYSTEM PROJECT



| Project ID: LS-028-Alt1                                                                                                                                                                                                                                                             |                           |     |      |           |              |            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-----|------|-----------|--------------|------------|
| Project Name: Ferry Terminal                                                                                                                                                                                                                                                        |                           |     |      |           |              |            |
| Project Location: City of Larkspur (37.9452, -122.5122)<br>(Project drains Caltrans right-of-way area as well)                                                                                                                                                                      |                           |     |      |           |              |            |
| Site/Device Information                                                                                                                                                                                                                                                             |                           |     |      |           |              |            |
| Proposed Device Type: Netting Devices (See page 2 for Alternatives)                                                                                                                                                                                                                 |                           |     |      |           |              |            |
| Project Description: Construction of an end-of-pipe netting system and concrete pad. The netting devices would be attached to the two 42-inch pipes discharging into the creek parallel to Larkspur Ferry Terminal Pathway.                                                         |                           |     |      |           |              |            |
| Site Specific Considerations: <ul style="list-style-type: none"><li>Pipe outfalls have existing flap gates</li><li>Location is tidal</li><li>Channel is privately owned</li></ul>                                                                                                   |                           |     |      |           |              |            |
| O&M Summary: May require creek drawdown for maintenance.                                                                                                                                                                                                                            |                           |     |      |           |              |            |
| Comments/Notes: GG transit owns the outfall and creek, so device installation would need to be coordinated. Permitting may present a challenge, however, upstream locations are problematic due to the high traffic along Sir Francis Drake. Upstream locations also miss Caltrans. |                           |     |      |           |              |            |
| Drainage Area and Trash Reduction Information                                                                                                                                                                                                                                       |                           |     |      |           |              |            |
| Total Drainage Area: 59 acres <ul style="list-style-type: none"><li>Larkspur – 54 acres</li><li>Caltrans Right-of-Way – 5 acres</li></ul>                                                                                                                                           |                           |     |      |           |              |            |
| Priority Land Use (PLU) Areas (acres): 41 acres                                                                                                                                                                                                                                     |                           |     |      |           |              |            |
| Public Agency                                                                                                                                                                                                                                                                       | Trash Generation Category |     |      |           | Non-PLU Area | Total Area |
|                                                                                                                                                                                                                                                                                     | Low                       | Mod | High | Very High |              |            |
| Larkspur                                                                                                                                                                                                                                                                            | 12                        | 29  | 0    | 0         | 13           | 54         |
| Caltrans                                                                                                                                                                                                                                                                            | --                        | --  | --   | --        | --           | 5          |
| 1-yr/1-hr Peak Flow (modeled): 24 cfs                                                                                                                                                                                                                                               |                           |     |      |           |              |            |



|                                   |             |
|-----------------------------------|-------------|
| Preliminary Cost Estimate         |             |
| Design/Permitting                 | \$173,000   |
| Construction                      | \$241,000   |
| Construction Contingency          | \$48,300    |
| Total Capital                     | \$523,000   |
| Annual Operation & Maintenance    | \$27,300/yr |
| Construction Cost per Acre of PLU | \$8,500/ac  |





CONCEPTUAL DESIGN - POTENTIAL FULL TRASH SYSTEM PROJECT



| Alternative Design Options |                                            |                                                                                                                                                    |                                                                                                            |                                                                   |                                                                                                                                           |                                           |                     |                                      |
|----------------------------|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|---------------------|--------------------------------------|
| Alternative Site ID        | Proposed Device Type                       | Project Description                                                                                                                                | Site Specific Considerations                                                                               | O&M Summary                                                       | Comments/Notes                                                                                                                            | Preliminary Cost Estimate                 |                     | Photos                               |
| LS-028-Alt2                | Hydrodynamic Separator (HDS) or Baffle Box | Construction of an underground hydrodynamic separator (HDS) or baffle box on the intersection of Larkspur Landing Circle and Si Francis Drake Blvd | <ul style="list-style-type: none"><li>High traffic area</li><li>Location does not treat Caltrans</li></ul> | Requires a vacuum-assisted truck and traffic control/lane closure | No daytime closures allowed on Sir Francis Drake Boulevard. Construction of HDS or baffle box unit at this location will be very complex. | Design/Permitting                         | \$81,000            | <div>Proposed Project Location</div> |
|                            |                                            |                                                                                                                                                    |                                                                                                            |                                                                   |                                                                                                                                           | Construction                              | \$360,000           |                                      |
|                            |                                            |                                                                                                                                                    |                                                                                                            |                                                                   |                                                                                                                                           | Construction Contingency                  | \$58,000            |                                      |
|                            |                                            |                                                                                                                                                    |                                                                                                            |                                                                   |                                                                                                                                           | <b>Total Capital</b>                      | <b>\$441,000</b>    |                                      |
|                            |                                            |                                                                                                                                                    |                                                                                                            |                                                                   |                                                                                                                                           | <b>Annual Operation &amp; Maintenance</b> | <b>\$23,400/yr</b>  |                                      |
|                            |                                            |                                                                                                                                                    |                                                                                                            |                                                                   |                                                                                                                                           | <b>Construction Cost per Acre of PLU</b>  | <b>\$8,700/ac</b>   |                                      |
| LS-028-Alt3                | Small Devices                              | This proposed alternative would entail the construction of approximately 40 inlet screens throughout the drainage area of this project.            | N/A                                                                                                        | Requires maintenance at many different locations.                 | Construction Contingency cost included in the overall Construction cost.                                                                  | Design/Permitting                         | \$20,000            | N/A                                  |
|                            |                                            |                                                                                                                                                    |                                                                                                            |                                                                   |                                                                                                                                           | Construction                              | \$100,000           |                                      |
|                            |                                            |                                                                                                                                                    |                                                                                                            |                                                                   |                                                                                                                                           | <b>Total Capital</b>                      | <b>\$120,000</b>    |                                      |
|                            |                                            |                                                                                                                                                    |                                                                                                            |                                                                   |                                                                                                                                           | <b>Annual Operation &amp; Maintenance</b> | <b>\$288,000/yr</b> |                                      |
|                            |                                            |                                                                                                                                                    |                                                                                                            |                                                                   |                                                                                                                                           | <b>Construction Cost per Acre of PLU</b>  | <b>\$2,900/ac</b>   |                                      |



CONCEPTUAL DESIGN - POTENTIAL FULL TRASH SYSTEM PROJECT



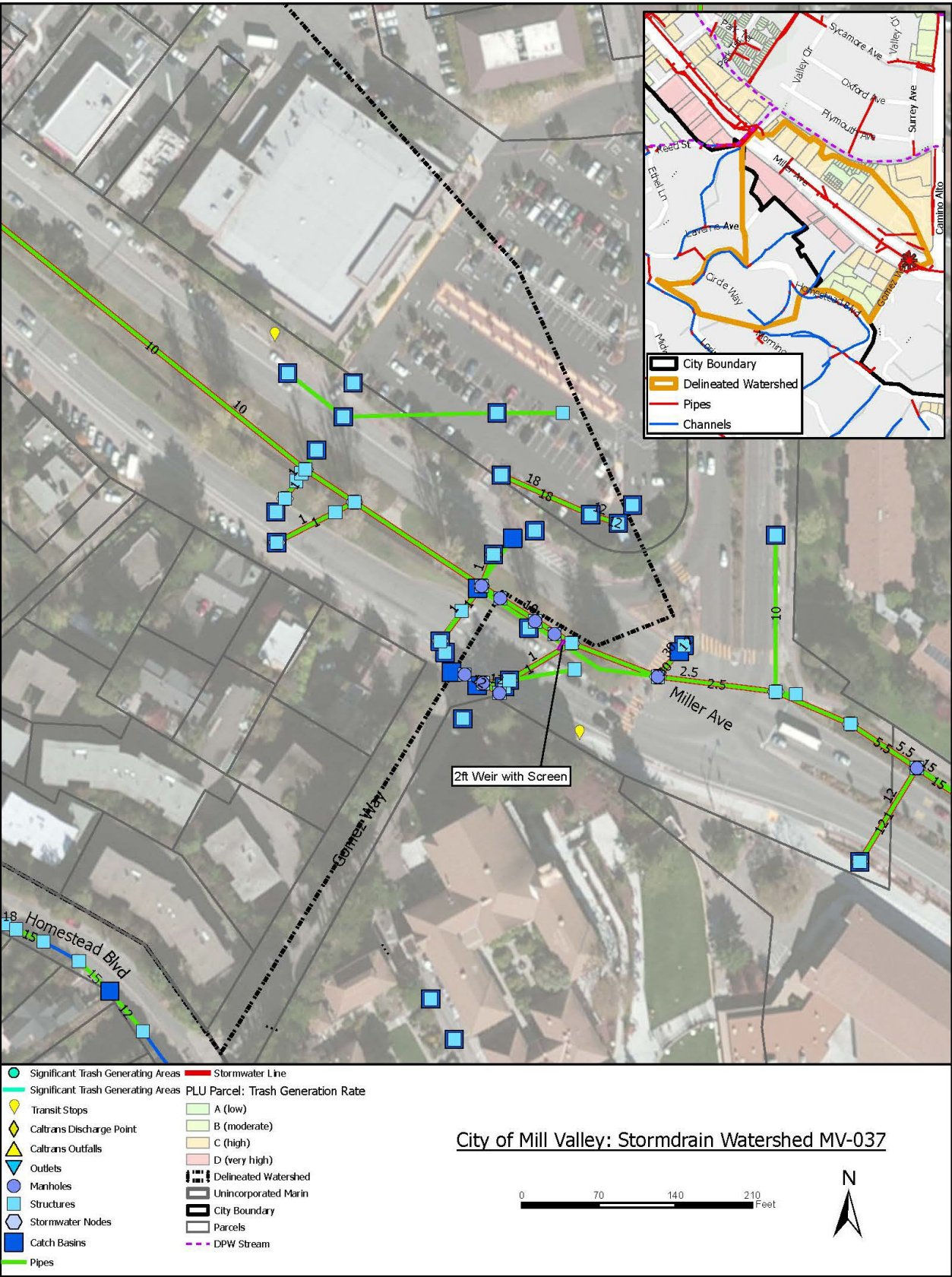
|                                                                                                      |
|------------------------------------------------------------------------------------------------------|
| <b>Project ID:</b> MV-037-AIt1                                                                       |
| <b>Project Name:</b> Miller Avenue                                                                   |
| <b>Project Location:</b> City of Mill Valley ( <a href="#">37.8931</a> , <a href="#">-122.5306</a> ) |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Site/Device Information</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Proposed Device Type:</b> Louvered Panel Gross Solid Removal Device (GSRD) (See page 2 for Alternatives)                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Project Description:</b> Construction of a custom louvered 5-mm GSRD panel screen across the concrete channel that is 10-foot in width. The panel screen will be sited directly downstream of where the underground box culvert daylights in the median on Miller Avenue. The custom panel screen will be roughly 24 inches in height and constructed with a 6-inch weir to allow overflow to occur and avoid flooding during large storms. Once the full capture, 1-yr, 1hr design flow is exceeded, flows in the channel will overflow the screen via the weir. |
| <b>Site Specific Considerations:</b> <ul style="list-style-type: none"><li>Location is likely tidal and known to flood</li><li>There is ample space in median for maintenance and operation of the louvered panel</li><li>Mature trees in the median would need to be avoided during construction and maintenance</li></ul>                                                                                                                                                                                                                                          |
| <b>O&amp;M Summary:</b> Device will be accessible via the turn lane on Miller Avenue.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Comments/Notes:</b> The project may require permission from the Water Board to install a device in the channel, however it is believed the channel is not a creek and can be considered part of the City's storm drain system. The project does not receive any drainage from Caltrans' jurisdictional area.                                                                                                                                                                                                                                                      |



| Drainage Area and Trash Reduction Information                                                                                                       |                           |     |      |           |              |            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-----|------|-----------|--------------|------------|
| <b>Total Drainage Area:</b> 29 acres <ul style="list-style-type: none"><li>Mill Valley – 29 acres</li><li>Caltrans Right-of-Way – 0 acres</li></ul> |                           |     |      |           |              |            |
| <b>Priority Land Use (PLU) Areas (acres):</b>                                                                                                       |                           |     |      |           |              |            |
| Public Agency                                                                                                                                       | Trash Generation Category |     |      |           | Non-PLU Area | Total Area |
|                                                                                                                                                     | Low                       | Mod | High | Very High |              |            |
| Mill Valley                                                                                                                                         | 1                         | 2   | 5    | 3         | 18           | 29         |
| Caltrans                                                                                                                                            | --                        | --  | --   | --        | --           | 0          |
| <b>1-yr/1-hr Peak Flow (modeled):</b> 8 cfs                                                                                                         |                           |     |      |           |              |            |

| Preliminary Cost Estimate                 |                   |
|-------------------------------------------|-------------------|
| Design/Permitting                         | \$32,000          |
| Construction                              | \$12,000          |
| Construction Contingency                  | \$2,400           |
| <b>Total Capital</b>                      | <b>\$56,000</b>   |
| <b>Annual Operation &amp; Maintenance</b> | <b>\$7,200/yr</b> |
| <b>Construction Cost per Acre of PLU</b>  | <b>\$2,200/ac</b> |





CONCEPTUAL DESIGN - POTENTIAL FULL TRASH SYSTEM PROJECT



| Alternative Design Options |                      |                                                                                                                                         |                              |                                                   |                                                                          |                                   |              |        |
|----------------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------------|---------------------------------------------------|--------------------------------------------------------------------------|-----------------------------------|--------------|--------|
| Alternative Site ID        | Proposed Device Type | Project Description                                                                                                                     | Site Specific Considerations | O&M Summary                                       | Comments/Notes                                                           | Preliminary Cost Estimate         |              | Photos |
| MV-037-Alt2                | Small Devices        | This proposed alternative would entail the construction of approximately 46 inlet screens throughout the drainage area of this project. | N/A                          | Requires maintenance at many different locations. | Construction Contingency cost included in the overall Construction cost. | Design/Permitting                 | \$23,000     | N/A    |
|                            |                      |                                                                                                                                         |                              |                                                   |                                                                          | Construction                      | \$115,000    |        |
|                            |                      |                                                                                                                                         |                              |                                                   |                                                                          | Total Capital                     | \$138,000    |        |
|                            |                      |                                                                                                                                         |                              |                                                   |                                                                          | Annual Operation & Maintenance    | \$331,000/yr |        |
|                            |                      |                                                                                                                                         |                              |                                                   |                                                                          | Construction Cost per Acre of PLU | \$12,800/ac  |        |



CONCEPTUAL DESIGN - POTENTIAL FULL TRASH SYSTEM PROJECT



|                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Project ID:</b> NV-029-Alt1                                                                                                        |
| <b>Project Name:</b> Commercial Drive                                                                                                 |
| <b>Project Location:</b> City of Novato ( <a href="#">38.0686, -122.5333</a> )<br>(Project drains Caltrans right-of-way area as well) |

|                                                                                                                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Site/Device Information</b>                                                                                                                                                                                                                                     |
| <b>Proposed Device Type:</b> Hydrodynamic Separator (HDS) (See page 2 for Alternatives)                                                                                                                                                                            |
| <b>Project Description:</b> Construction of an underground hydrodynamic separator (HDS) unit. The large device would be installed within an existing 48-inch pipe located underneath Commercial Boulevard at the edge of private property.                         |
| <b>Site Specific Considerations:</b> <ul style="list-style-type: none"><li>Location is close to private property</li><li>Lack of information on upstream drainage area</li><li>Some Caltrans area may drain directly to creek and not through the device</li></ul> |
| <b>O&amp;M Summary:</b> Requires a vacuum-assisted truck                                                                                                                                                                                                           |
| <b>Comments/Notes:</b> Project will likely be considered a CEQA Categorical Exemption.                                                                                                                                                                             |



| Drainage Area and Trash Reduction Information                                                                                                   |                           |     |      |           |              |            |
|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-----|------|-----------|--------------|------------|
| <b>Total Drainage Area:</b> 25 acres <ul style="list-style-type: none"><li>Novato – 13 acres</li><li>Caltrans Right-of-Way – 12 acres</li></ul> |                           |     |      |           |              |            |
| <b>Priority Land Use (PLU) Areas (acres):</b> 4 acres                                                                                           |                           |     |      |           |              |            |
| Public Agency                                                                                                                                   | Trash Generation Category |     |      |           | Non-PLU Area | Total Area |
|                                                                                                                                                 | Low                       | Mod | High | Very High |              |            |
| Novato                                                                                                                                          | 0                         | 2   | 2    | 0         | 9            | 13         |
| Caltrans                                                                                                                                        | --                        | --  | --   | --        | --           | 12         |
| <b>1-yr/1-hr Peak Flow (modeled):</b> 10 cfs                                                                                                    |                           |     |      |           |              |            |

| Preliminary Cost Estimate                 |                    |
|-------------------------------------------|--------------------|
| Design/Permitting                         | \$52,000           |
| Construction                              | \$116,700          |
| Construction Contingency                  | \$23,300           |
| <b>Total Capital</b>                      | <b>\$222,000</b>   |
| <b>Annual Operation &amp; Maintenance</b> | <b>\$23,400/yr</b> |
| <b>Construction Cost per Acre of PLU</b>  | <b>\$39,000/ac</b> |





CONCEPTUAL DESIGN - POTENTIAL FULL TRASH SYSTEM PROJECT



| Alternative Design Options |                      |                                                                                                                                        |                              |                                                   |                                                                          |                                   |             |        |
|----------------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------|---------------------------------------------------|--------------------------------------------------------------------------|-----------------------------------|-------------|--------|
| Alternative Site ID        | Proposed Device Type | Project Description                                                                                                                    | Site Specific Considerations | O&M Summary                                       | Comments/Notes                                                           | Preliminary Cost Estimate         |             | Photos |
| NV-029-Alt2                | Small Devices        | This proposed alternative would entail the construction of approximately 7 inlet screens throughout the drainage area of this project. | N/A                          | Requires maintenance at many different locations. | Construction Contingency cost included in the overall Construction cost. | Design/Permitting                 | \$10,500    | N/A    |
|                            |                      |                                                                                                                                        |                              |                                                   |                                                                          | Construction                      | \$17,500    |        |
|                            |                      |                                                                                                                                        |                              |                                                   |                                                                          | Total Capital                     | \$28,000    |        |
|                            |                      |                                                                                                                                        |                              |                                                   |                                                                          | Annual Operation & Maintenance    | \$50,400/yr |        |
|                            |                      |                                                                                                                                        |                              |                                                   |                                                                          | Construction Cost per Acre of PLU | \$6,440/ac  |        |



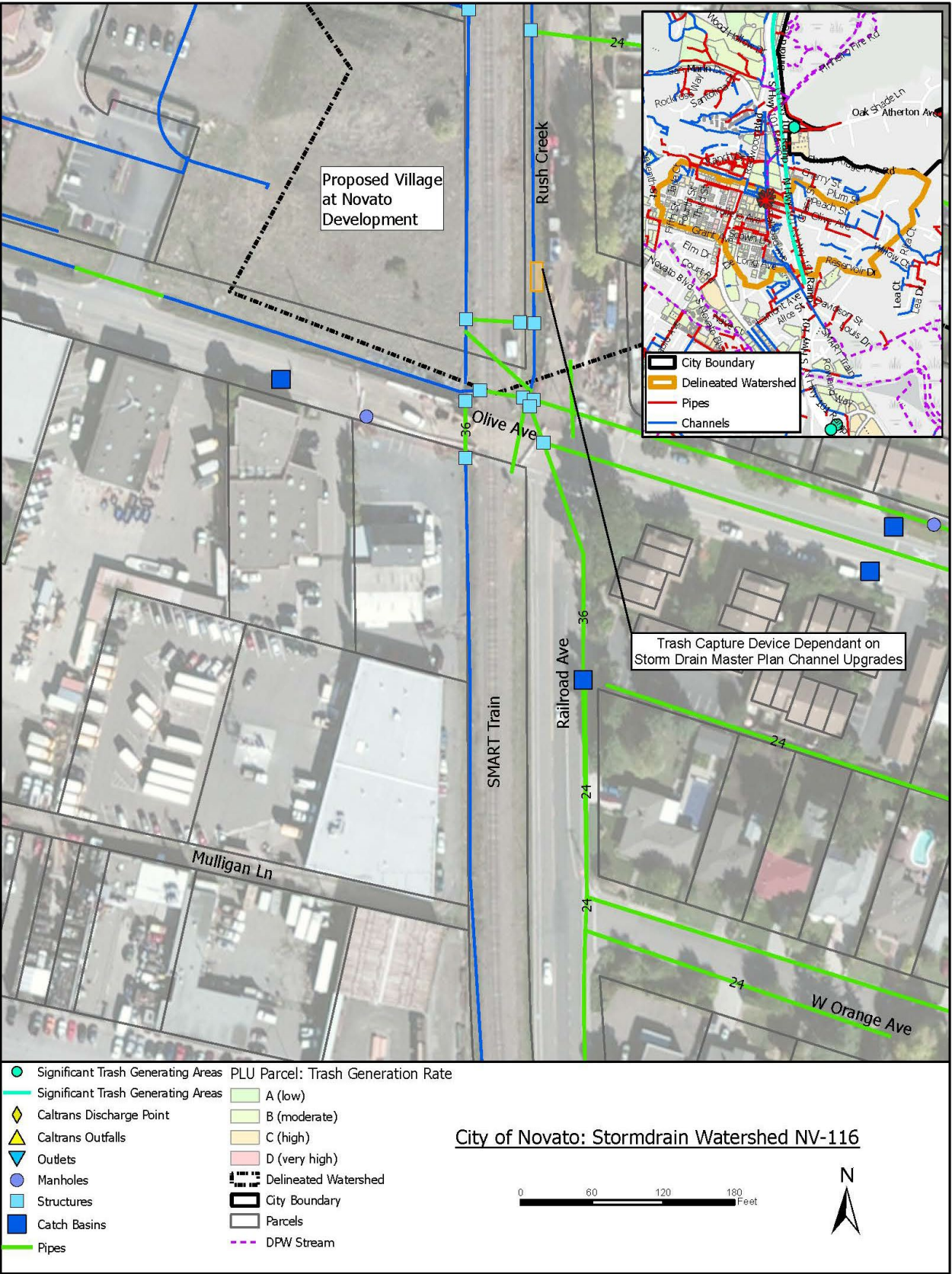
CONCEPTUAL DESIGN - POTENTIAL FULL TRASH SYSTEM PROJECT



| Project ID: NV-116-Alt1                                                                                                                                                                                                                                                                                                                                                                                                    |                           |     |      |           |              |            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-----|------|-----------|--------------|------------|
| Project Name: Olive Avenue                                                                                                                                                                                                                                                                                                                                                                                                 |                           |     |      |           |              |            |
| Project Location: City of Novato (38.1106, -122.5663)<br>(Project drains Caltrans right-of-way area as well)                                                                                                                                                                                                                                                                                                               |                           |     |      |           |              |            |
| Site/Device Information                                                                                                                                                                                                                                                                                                                                                                                                    |                           |     |      |           |              |            |
| Proposed Device Type: Louvered Panel Gross Solid Removal Device (GSRD) (See page 2 for Alternatives)                                                                                                                                                                                                                                                                                                                       |                           |     |      |           |              |            |
| Project Description: Installation of a louvered 5-mm screen in the North Railroad Ditch (NRR Ditch). The NRR Ditch runs parallel to the SMART tracks and Mulligan Lane. The ditch is 15-feet wide and 5-feet deep.                                                                                                                                                                                                         |                           |     |      |           |              |            |
| Site Specific Considerations: <ul style="list-style-type: none"><li>Location is adjacent to SMART tracks</li><li>NRR Ditch has capacity issue and is subject to backwater</li></ul>                                                                                                                                                                                                                                        |                           |     |      |           |              |            |
| O&M Summary: N/A                                                                                                                                                                                                                                                                                                                                                                                                           |                           |     |      |           |              |            |
| Comments/Notes: Permitting for this project may be expensive. The City of Novato's Storm Drain Master Plan Report identifies the NRR Ditch as a location for a storm drainage improvement project. The improvement project would involve the widening of the ditch and installation of a 5-feet by 3-feet precast RC Box Culvert. The capacity of this ditch must be increased prior to trash capture device installation. |                           |     |      |           |              |            |
| Drainage Area and Trash Reduction Information                                                                                                                                                                                                                                                                                                                                                                              |                           |     |      |           |              |            |
| Total Drainage Area: 367 acres <ul style="list-style-type: none"><li>Novato – 341 acres</li><li>Caltrans Right-of-Way – 26 acres</li></ul>                                                                                                                                                                                                                                                                                 |                           |     |      |           |              |            |
| Priority Land Use (PLU) Areas (acres): 77 acres                                                                                                                                                                                                                                                                                                                                                                            |                           |     |      |           |              |            |
| Public Agency                                                                                                                                                                                                                                                                                                                                                                                                              | Trash Generation Category |     |      |           | Non-PLU Area | Total Area |
|                                                                                                                                                                                                                                                                                                                                                                                                                            | Low                       | Mod | High | Very High |              |            |
| Novato                                                                                                                                                                                                                                                                                                                                                                                                                     | 0                         | 53  | 18   | 6         | 264          | 341        |
| Caltrans                                                                                                                                                                                                                                                                                                                                                                                                                   | --                        | --  | --   | --        | --           | 26         |
| 1-yr/1-hr Peak Flow (modeled): 90 cfs                                                                                                                                                                                                                                                                                                                                                                                      |                           |     |      |           |              |            |

Proposed Project Location

Proposed Project Location





CONCEPTUAL DESIGN - POTENTIAL FULL TRASH SYSTEM PROJECT



| Alternative Design Options |                      |                                                                                                                                          |                              |                                                   |                                                                          |                                   |                |        |
|----------------------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|---------------------------------------------------|--------------------------------------------------------------------------|-----------------------------------|----------------|--------|
| Alternative Site ID        | Proposed Device Type | Project Description                                                                                                                      | Site Specific Considerations | O&M Summary                                       | Comments/Notes                                                           | Preliminary Cost Estimate         |                | Photos |
| NV-116-Alt2                | Small Devices        | This proposed alternative would entail the construction of approximately 237 inlet screens throughout the drainage area of this project. | N/A                          | Requires maintenance at many different locations. | Construction Contingency cost included in the overall Construction cost. | Design/Permitting                 | \$119,000      | N/A    |
|                            |                      |                                                                                                                                          |                              |                                                   |                                                                          | Construction                      | \$593,000      |        |
|                            |                      |                                                                                                                                          |                              |                                                   |                                                                          | Total Capital                     | \$712,000      |        |
|                            |                      |                                                                                                                                          |                              |                                                   |                                                                          | Annual Operation & Maintenance    | \$1,710,000/yr |        |
|                            |                      |                                                                                                                                          |                              |                                                   |                                                                          | Construction Cost per Acre of PLU | \$8,860 /ac    |        |

CONCEPTUAL DESIGN - POTENTIAL FULL TRASH SYSTEM PROJECT

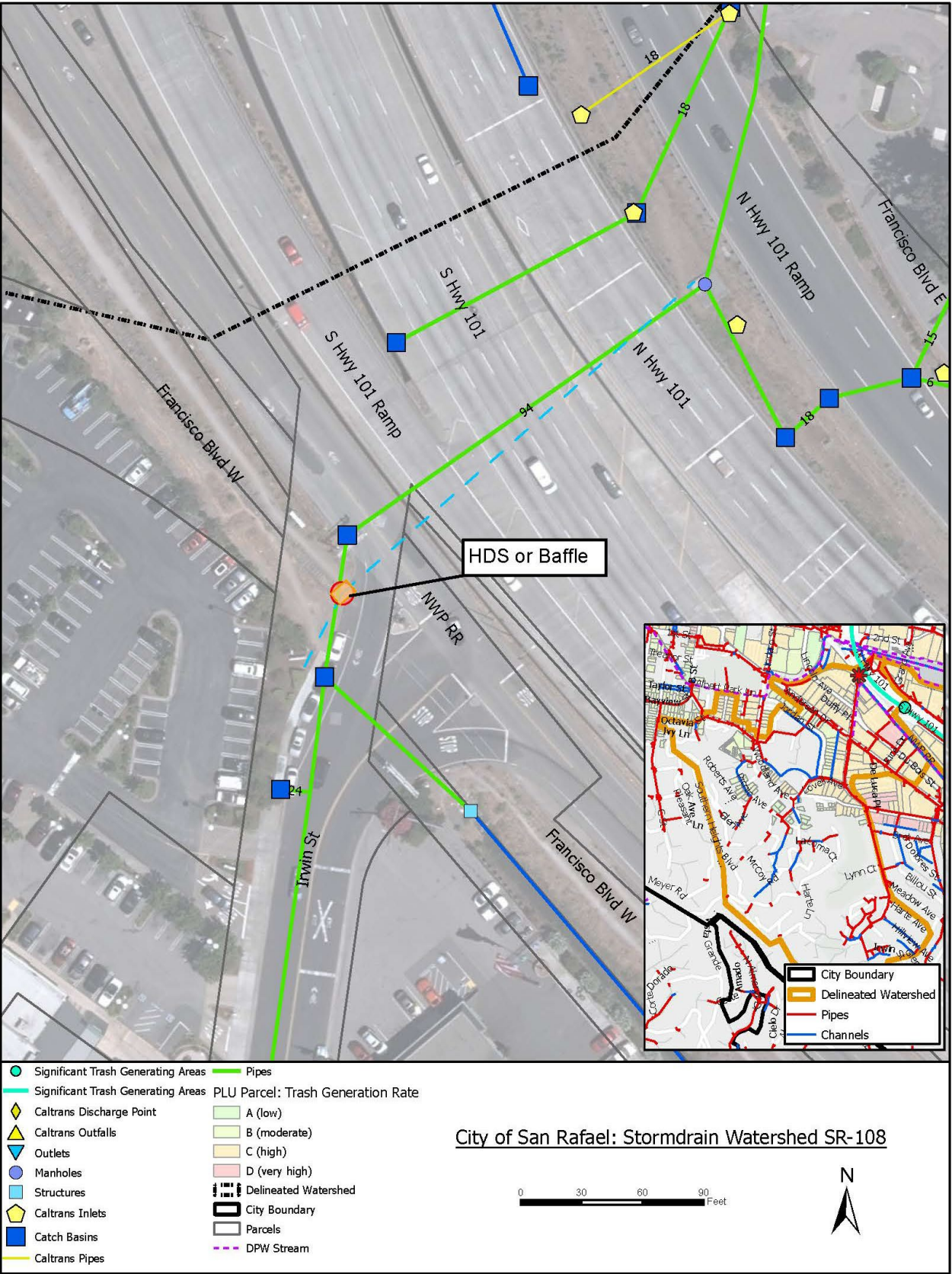


| Project ID: SR-108-Alt1                                                                                                                                                                                                                                                                                                                                                                  |                           |     |      |           |              |            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-----|------|-----------|--------------|------------|
| Project Name: N Francisco Pump Station                                                                                                                                                                                                                                                                                                                                                   |                           |     |      |           |              |            |
| Project Location: City of San Rafael (37.9696, -122.5208)<br>(Project drains Caltrans right-of-way area as well)                                                                                                                                                                                                                                                                         |                           |     |      |           |              |            |
| Site/Device Information                                                                                                                                                                                                                                                                                                                                                                  |                           |     |      |           |              |            |
| Proposed Device Type: Hydrodynamic Separator (HDS) or Baffle Box (See page 2 for Alternatives)                                                                                                                                                                                                                                                                                           |                           |     |      |           |              |            |
| Project Description: Construction of a hydrodynamic separator (HDS) or baffle box unit in Francisco Boulevard West near SMART tracks.                                                                                                                                                                                                                                                    |                           |     |      |           |              |            |
| Site Specific Considerations: <ul style="list-style-type: none"><li>Adjacent to SMART tracks</li><li>Pump station downstream</li></ul>                                                                                                                                                                                                                                                   |                           |     |      |           |              |            |
| O&M Summary: Requires a vacuum-assisted truck                                                                                                                                                                                                                                                                                                                                            |                           |     |      |           |              |            |
| Comments/Notes: Location at pump station downstream was deemed infeasible and moved to other side of the freeway. As a result, no Caltrans area will be treated by Alt1 proposed device. Project will require coordination with the Sonoma-Marin Area Rail Transit (SMART). Initially, a device at the pump station was proposed, however, a device at the pump station is not feasible. |                           |     |      |           |              |            |
| Drainage Area and Trash Reduction Information                                                                                                                                                                                                                                                                                                                                            |                           |     |      |           |              |            |
| Total Drainage Area: 266 acres <ul style="list-style-type: none"><li>San Rafael – 266 acres</li><li>Caltrans Right-of-Way – 0 acres</li></ul>                                                                                                                                                                                                                                            |                           |     |      |           |              |            |
| Priority Land Use (PLU) Areas (acres): 55 acres                                                                                                                                                                                                                                                                                                                                          |                           |     |      |           |              |            |
| Public Agency                                                                                                                                                                                                                                                                                                                                                                            | Trash Generation Category |     |      |           | Non-PLU Area | Total Area |
|                                                                                                                                                                                                                                                                                                                                                                                          | Low                       | Mod | High | Very High |              |            |
| San Rafael                                                                                                                                                                                                                                                                                                                                                                               | 0                         | 10  | 40   | 4         | 212          | 266        |
| Caltrans                                                                                                                                                                                                                                                                                                                                                                                 | --                        | --  | --   | --        | --           | 0          |
| 1-yr/1-hr Peak Flow (modeled): 97 cfs                                                                                                                                                                                                                                                                                                                                                    |                           |     |      |           |              |            |

Proposed Project Location

Proposed Project Location

|                                   |             |
|-----------------------------------|-------------|
| Preliminary Cost Estimate         |             |
| Design/Permitting                 | \$279,000   |
| Construction                      | \$1,160,000 |
| Construction Contingency          | \$232,000   |
| Total Capital                     | \$1,960,000 |
| Annual Operation & Maintenance    | \$23,400/yr |
| Construction Cost per Acre of PLU | \$30,800/ac |





CONCEPTUAL DESIGN - POTENTIAL FULL TRASH SYSTEM PROJECT



| Alternative Design Options |                      |                                                                                                                                          |                              |                                                   |                                                                          |                                   |                |        |
|----------------------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|---------------------------------------------------|--------------------------------------------------------------------------|-----------------------------------|----------------|--------|
| Alternative Site ID        | Proposed Device Type | Project Description                                                                                                                      | Site Specific Considerations | O&M Summary                                       | Comments/Notes                                                           | Preliminary Cost Estimate         |                | Photos |
| SR-108-Alt2                | Small Devices        | This proposed alternative would entail the construction of approximately 206 inlet screens throughout the drainage area of this project. | N/A                          | Requires maintenance at many different locations. | Construction Contingency cost included in the overall Construction cost. | Design/ Permitting                | \$103,000      | N/A    |
|                            |                      |                                                                                                                                          |                              |                                                   |                                                                          | Construction                      | \$515,000      |        |
|                            |                      |                                                                                                                                          |                              |                                                   |                                                                          | Total Capital                     | \$618,000      |        |
|                            |                      |                                                                                                                                          |                              |                                                   |                                                                          | Annual Operation & Maintenance    | \$1,480,000/yr |        |
|                            |                      |                                                                                                                                          |                              |                                                   |                                                                          | Construction Cost per Acre of PLU | \$11,300 /ac   |        |

CONCEPTUAL DESIGN - POTENTIAL FULL TRASH SYSTEM PROJECT



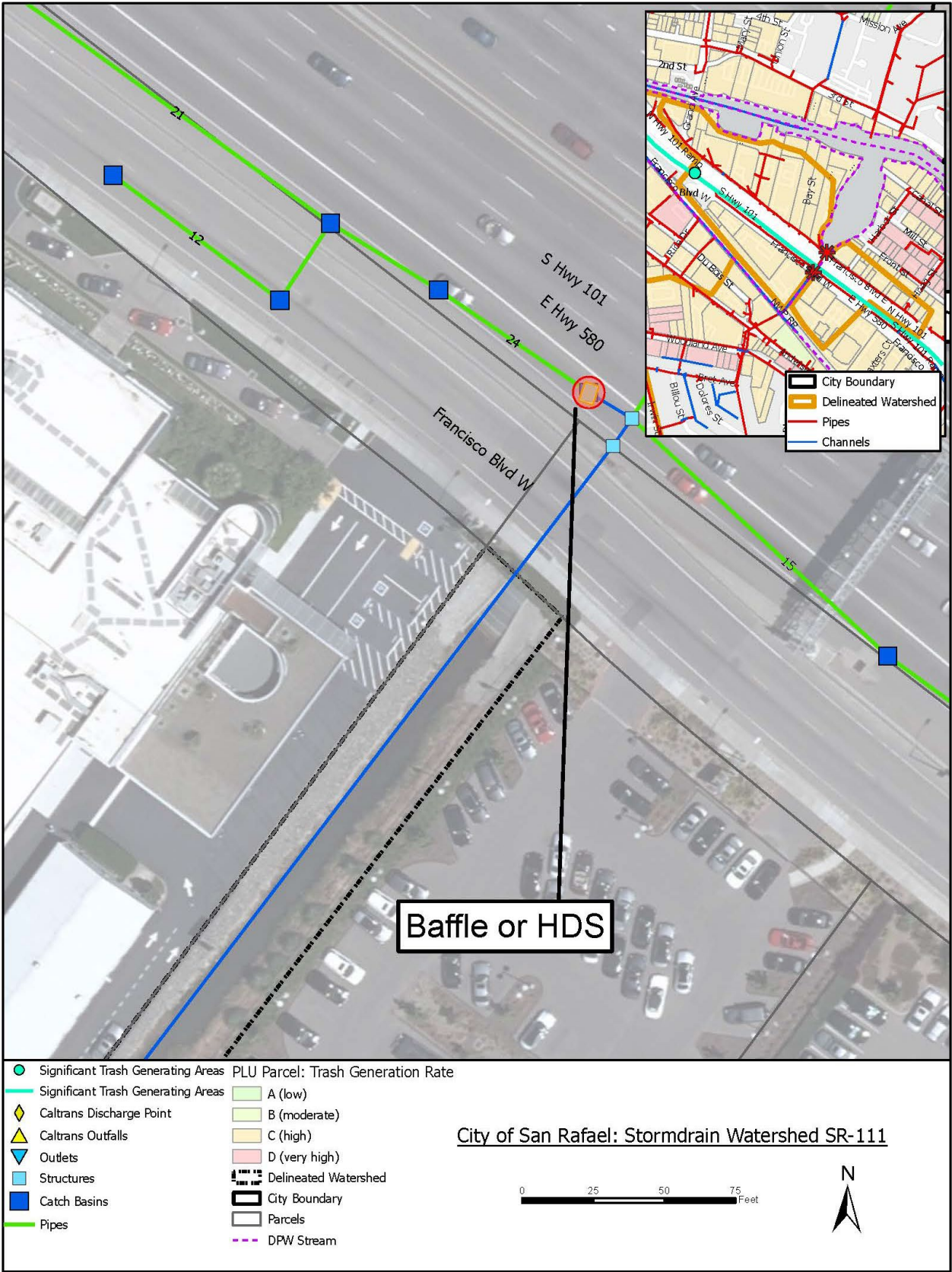
|                                                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Project ID:</b> SR-111-Alt1                                                                                                            |
| <b>Project Name:</b> Sir Francis Drake W                                                                                                  |
| <b>Project Location:</b> City of San Rafael ( <a href="#">37.9650, -122.5151</a> )<br>(Project drains Caltrans right-of-way area as well) |

|                                                                                                                                                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Site/Device Information</b>                                                                                                                                                                                                             |
| <b>Proposed Device Type:</b> Hydrodynamic Separator (HDS) or Baffle Box (See page 2 for Alternatives)                                                                                                                                      |
| <b>Project Description:</b> Construction of an underground hydrodynamic separator (HDS) or baffle box unit on the system that runs along E Highway 101 to capture runoff from the freeway and the surrounding Si Francis Drake properties. |
| <b>Site Specific Considerations:</b> <ul style="list-style-type: none"><li>Located in Caltrans ROW</li></ul>                                                                                                                               |
| <b>O&amp;M Summary:</b> Requires a vacuum-assisted truck and traffic control/lane closure                                                                                                                                                  |
| <b>Comments/Notes:</b> Site SR-143 should be a priority over this project location as SR-143 covers this area and more PLU.                                                                                                                |

| Drainage Area and Trash Reduction Information                                                                                                     |                           |     |      |           |              |            |
|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-----|------|-----------|--------------|------------|
| <b>Total Drainage Area:</b> 16 acres <ul style="list-style-type: none"><li>San Rafael – 8 acres</li><li>Caltrans Right-of-Way – 8 acres</li></ul> |                           |     |      |           |              |            |
| <b>Priority Land Use (PLU) Areas (acres):</b> 8 acres                                                                                             |                           |     |      |           |              |            |
| Public Agency                                                                                                                                     | Trash Generation Category |     |      |           | Non-PLU Area | Total Area |
|                                                                                                                                                   | Low                       | Mod | High | Very High |              |            |
| San Rafael                                                                                                                                        | 0                         | 0   | 8    | 0         | 0            | 8          |
| Caltrans                                                                                                                                          | --                        | --  | --   | --        | --           | 8          |
| <b>1-yr/1-hr Peak Flow (modeled):</b> 8 cfs                                                                                                       |                           |     |      |           |              |            |



| Preliminary Cost Estimate                 |                    |
|-------------------------------------------|--------------------|
| Design/Permitting                         | \$47,000           |
| Construction                              | \$95,500           |
| Construction Contingency                  | \$19,100           |
| <b>Total Capital</b>                      | <b>\$181,000</b>   |
| <b>Annual Operation &amp; Maintenance</b> | <b>\$23,400/yr</b> |
| <b>Construction Cost per Acre of PLU</b>  | <b>\$16,900/ac</b> |





CONCEPTUAL DESIGN - POTENTIAL FULL TRASH SYSTEM PROJECT



| Alternative Design Options |                      |                                                                                                                                        |                              |                                                   |                                                                                                                   |                                   |             |        |
|----------------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|-----------------------------------|-------------|--------|
| Alternative Site ID        | Proposed Device Type | Project Description                                                                                                                    | Site Specific Considerations | O&M Summary                                       | Comments/Notes                                                                                                    | Preliminary Cost Estimate         |             | Photos |
| SR-111-Alt2                | Small Devices        | This proposed alternative would entail the construction of approximately 4 inlet screens throughout the drainage area of this project. | N/A                          | Requires maintenance at many different locations. | Construction Contingency cost included in the overall Construction cost. Most inlets will be within Caltrans ROW. | Design/ Permitting                | \$6,000     | N/A    |
|                            |                      |                                                                                                                                        |                              |                                                   |                                                                                                                   | Construction                      | \$10,000    |        |
|                            |                      |                                                                                                                                        |                              |                                                   |                                                                                                                   | Total Capital                     | \$16,000    |        |
|                            |                      |                                                                                                                                        |                              |                                                   |                                                                                                                   | Annual Operation & Maintenance    | \$28,800/yr |        |
|                            |                      |                                                                                                                                        |                              |                                                   |                                                                                                                   | Construction Cost per Acre of PLU | \$6,200/ac  |        |



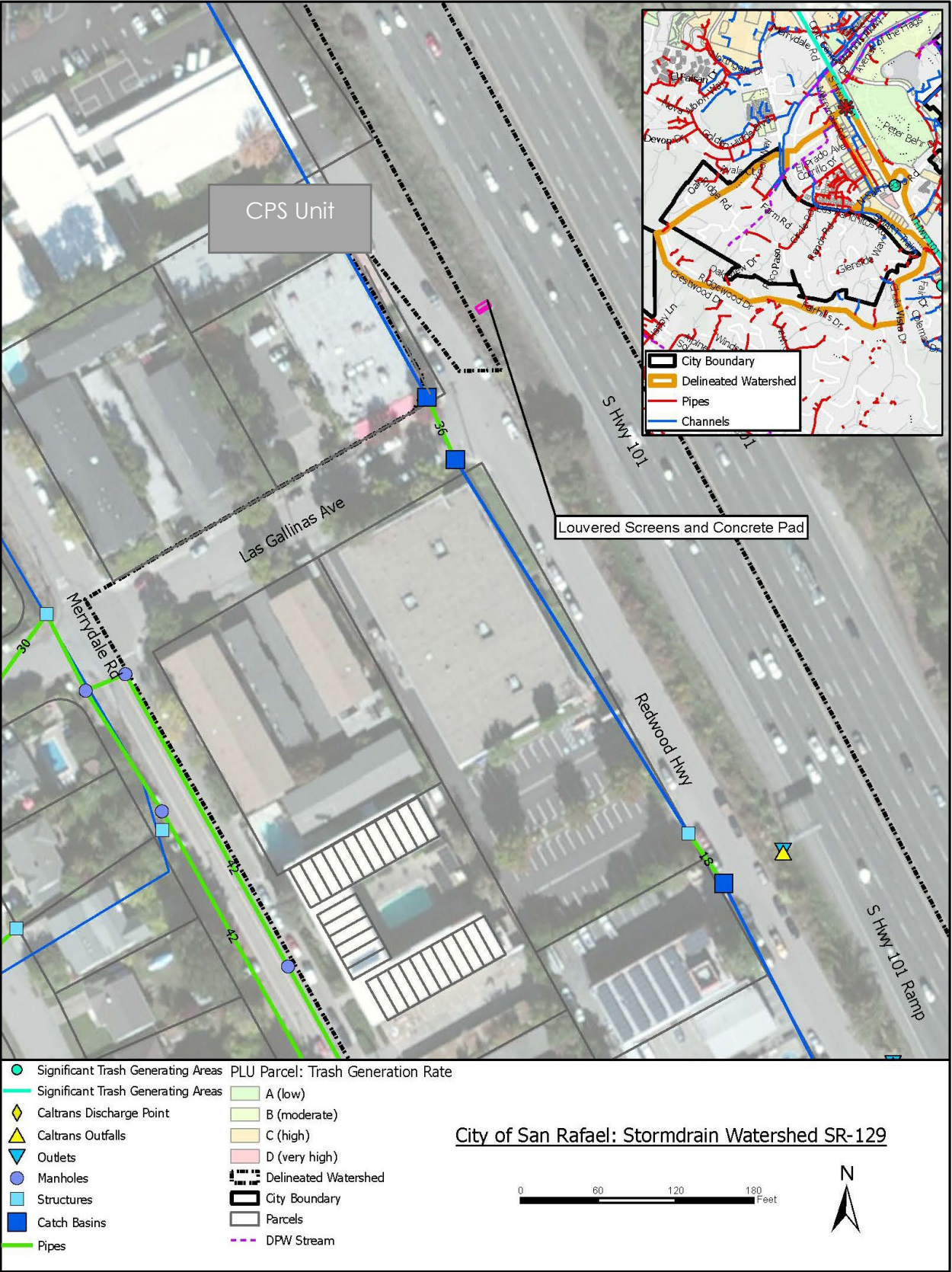
CONCEPTUAL DESIGN - POTENTIAL FULL TRASH SYSTEM PROJECT



| Project ID: SR-129-A1#1                                                                                                                                                                                                                                                                             |                           |     |      |           |              |            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-----|------|-----------|--------------|------------|
| Project Name: Redwood Highway                                                                                                                                                                                                                                                                       |                           |     |      |           |              |            |
| Project Location: City of San Rafael (37.9989, -122.5362)<br>(Project drains Caltrans right-of-way area as well)                                                                                                                                                                                    |                           |     |      |           |              |            |
| Site/Device Information                                                                                                                                                                                                                                                                             |                           |     |      |           |              |            |
| Proposed Device Type: Louvered Panel Gross Solid Removal Device (GSRD) (See page 2 for Alternatives) and one CPS unit                                                                                                                                                                               |                           |     |      |           |              |            |
| Project Description: Installation of a louvered 5-mm screen within the open ditch parallel to Redwood Highway and one CPS unit at the catch basin in front of restaurant.                                                                                                                           |                           |     |      |           |              |            |
| Site Specific Considerations: <ul style="list-style-type: none"><li>Telephone poles overhead</li><li>Permitting could be slightly more than on underground storm drain system</li></ul>                                                                                                             |                           |     |      |           |              |            |
| O&M Summary: N/A                                                                                                                                                                                                                                                                                    |                           |     |      |           |              |            |
| Comments/Notes: Project will require coordination with the SF Bay Regional Water Board (Water Board), which oversees water quality certification programs for creeks/channels in the Bay Area. The project is likely dependent on permission from the Water Board to install a screen in a channel. |                           |     |      |           |              |            |
| Drainage Area and Trash Reduction Information                                                                                                                                                                                                                                                       |                           |     |      |           |              |            |
| Total Drainage Area: 44 acres <ul style="list-style-type: none"><li>San Rafael – 24 acres</li><li>Caltrans Right-of-Way – 20 acres</li></ul>                                                                                                                                                        |                           |     |      |           |              |            |
| Priority Land Use (PLU) Areas (acres): 18 acres                                                                                                                                                                                                                                                     |                           |     |      |           |              |            |
| Public Agency                                                                                                                                                                                                                                                                                       | Trash Generation Category |     |      |           | Non-PLU Area | Total Area |
|                                                                                                                                                                                                                                                                                                     | Low                       | Mod | High | Very High |              |            |
| San Rafael                                                                                                                                                                                                                                                                                          | 0                         | 5   | 11   | 2         | 6            | 24         |
| Caltrans                                                                                                                                                                                                                                                                                            | --                        | --  | --   | --        | --           | 20         |
| 1-yr/1-hr Peak Flow (modeled): 19 cfs                                                                                                                                                                                                                                                               |                           |     |      |           |              |            |

Proposed Project Location

Proposed Project Location





CONCEPTUAL DESIGN - POTENTIAL FULL TRASH SYSTEM PROJECT



| Alternative Design Options |                      |                                                                                                                                         |                              |                                                   |                                                                          |                                   |              |        |
|----------------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------------|---------------------------------------------------|--------------------------------------------------------------------------|-----------------------------------|--------------|--------|
| Alternative Site ID        | Proposed Device Type | Project Description                                                                                                                     | Site Specific Considerations | O&M Summary                                       | Comments/Notes                                                           | Preliminary Cost Estimate         |              | Photos |
| SR-129-Alt2                | Small Devices        | This proposed alternative would entail the construction of approximately 26 inlet screens throughout the drainage area of this project. | N/A                          | Requires maintenance at many different locations. | Construction Contingency cost included in the overall Construction cost. | Design/ Permitting                | \$13,000     | N/A    |
|                            |                      |                                                                                                                                         |                              |                                                   |                                                                          | Construction                      | \$65,000     |        |
|                            |                      |                                                                                                                                         |                              |                                                   |                                                                          | Total Capital                     | \$78,000     |        |
|                            |                      |                                                                                                                                         |                              |                                                   |                                                                          | Annual Operation & Maintenance    | \$187,000/yr |        |
|                            |                      |                                                                                                                                         |                              |                                                   |                                                                          | Construction Cost per Acre of PLU | \$4,300 /ac  |        |

CONCEPTUAL DESIGN - POTENTIAL FULL TRASH SYSTEM PROJECT



|                                                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Project ID:</b> SR-143-Alt1                                                                                                            |
| <b>Project Name:</b> Rossi Pump Station                                                                                                   |
| <b>Project Location:</b> City of San Rafael ( <a href="#">37.9655, -122.5147</a> )<br>(Project drains Caltrans right-of-way area as well) |

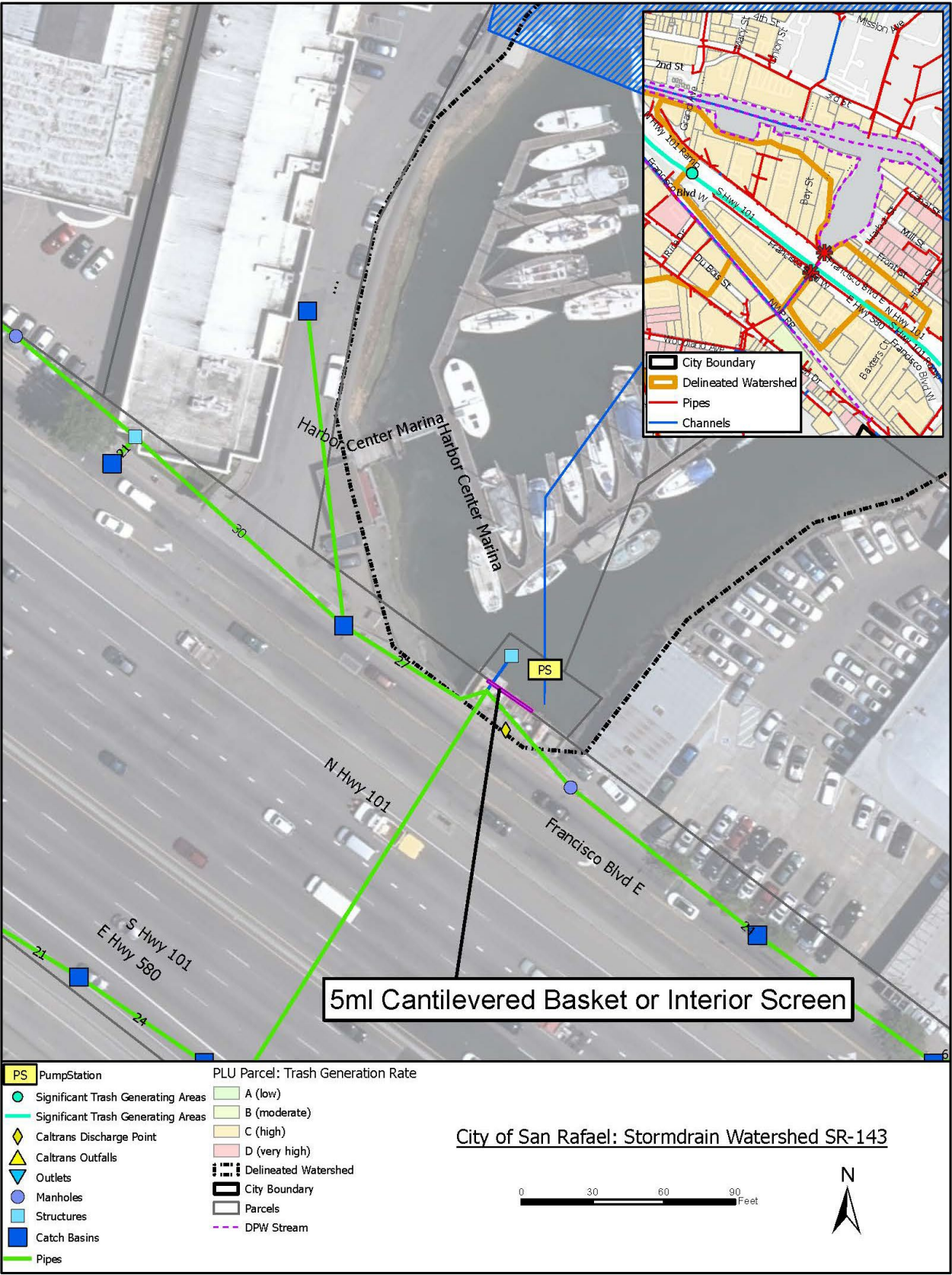
|                                                                                                                                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Site/Device Information</b>                                                                                                                                                                                                                                              |
| <b>Proposed Device Type:</b> Custom Screen (See page 2 for Alternatives)                                                                                                                                                                                                    |
| <b>Project Description:</b> Construction of a custom cantilevered 5-mm screen attached to the bottom 6-inches of the existing rack on the outfall structure of the Rossi Pump Station. Three screens will be needed for each of the three outfalls at Rossi Pump Station.   |
| <b>Site Specific Considerations:</b> <ul style="list-style-type: none"><li>Pump station outfalls hang over San Rafael Creek</li><li>Risk of pump back-up</li></ul>                                                                                                          |
| <b>O&amp;M Summary:</b> Pumps will need to be turned off during maintenance                                                                                                                                                                                                 |
| <b>Comments/Notes:</b> There are three pumps with individual boxes and outfalls, this project could be phased so that one outfall structure is configured with 5mm screen to test the functionality of the screening. In addition pump station operations could be changed. |

Proposed Project Location

Pump at Rossi Pump Station

| Drainage Area and Trash Reduction Information                                                                                                       |                           |     |      |           |              |            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-----|------|-----------|--------------|------------|
| <b>Total Drainage Area:</b> 52 acres <ul style="list-style-type: none"><li>San Rafael – 36 acres</li><li>Caltrans Right-of-Way – 16 acres</li></ul> |                           |     |      |           |              |            |
| <b>Priority Land Use (PLU) Areas (acres):</b> 35 acres                                                                                              |                           |     |      |           |              |            |
| Public Agency                                                                                                                                       | Trash Generation Category |     |      |           | Non-PLU Area | Total Area |
|                                                                                                                                                     | Low                       | Mod | High | Very High |              |            |
| San Rafael                                                                                                                                          | 0                         | 0   | 35   | 0         | 1            | 36         |
| Caltrans                                                                                                                                            | --                        | --  | --   | --        | --           | 16         |
| <b>1-yr/1-hr Peak Flow (modeled):</b> 25 cfs                                                                                                        |                           |     |      |           |              |            |

| Preliminary Cost Estimate                 |                   |
|-------------------------------------------|-------------------|
| Design/Permitting                         | \$69,000          |
| Construction                              | \$100,000         |
| Construction Contingency                  | \$20,000          |
| <b>Total Capital</b>                      | <b>\$349,000</b>  |
| <b>Annual Operation &amp; Maintenance</b> | <b>\$9,600/yr</b> |
| <b>Construction Cost per Acre of PLU</b>  | <b>\$8,000/ac</b> |





CONCEPTUAL DESIGN - POTENTIAL FULL TRASH SYSTEM PROJECT



| Alternative Design Options |                      |                                                                                                                                         |                              |                                                   |                                                                          |                                   |              |        |
|----------------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------------|---------------------------------------------------|--------------------------------------------------------------------------|-----------------------------------|--------------|--------|
| Alternative Site ID        | Proposed Device Type | Project Description                                                                                                                     | Site Specific Considerations | O&M Summary                                       | Comments/Notes                                                           | Preliminary Cost Estimate         |              | Photos |
| SR-143-Alt2                | Small Devices        | This proposed alternative would entail the construction of approximately 80 inlet screens throughout the drainage area of this project. | N/A                          | Requires maintenance at many different locations. | Construction Contingency cost included in the overall Construction cost. | Design/ Permitting                | \$40,000     | N/A    |
|                            |                      |                                                                                                                                         |                              |                                                   |                                                                          | Construction                      | \$200,000    |        |
|                            |                      |                                                                                                                                         |                              |                                                   |                                                                          | Total Capital                     | \$240,000    |        |
|                            |                      |                                                                                                                                         |                              |                                                   |                                                                          | Annual Operation & Maintenance    | \$576,000/yr |        |
|                            |                      |                                                                                                                                         |                              |                                                   |                                                                          | Construction Cost per Acre of PLU | \$6,820/ac   |        |



CONCEPTUAL DESIGN - POTENTIAL FULL TRASH SYSTEM PROJECT

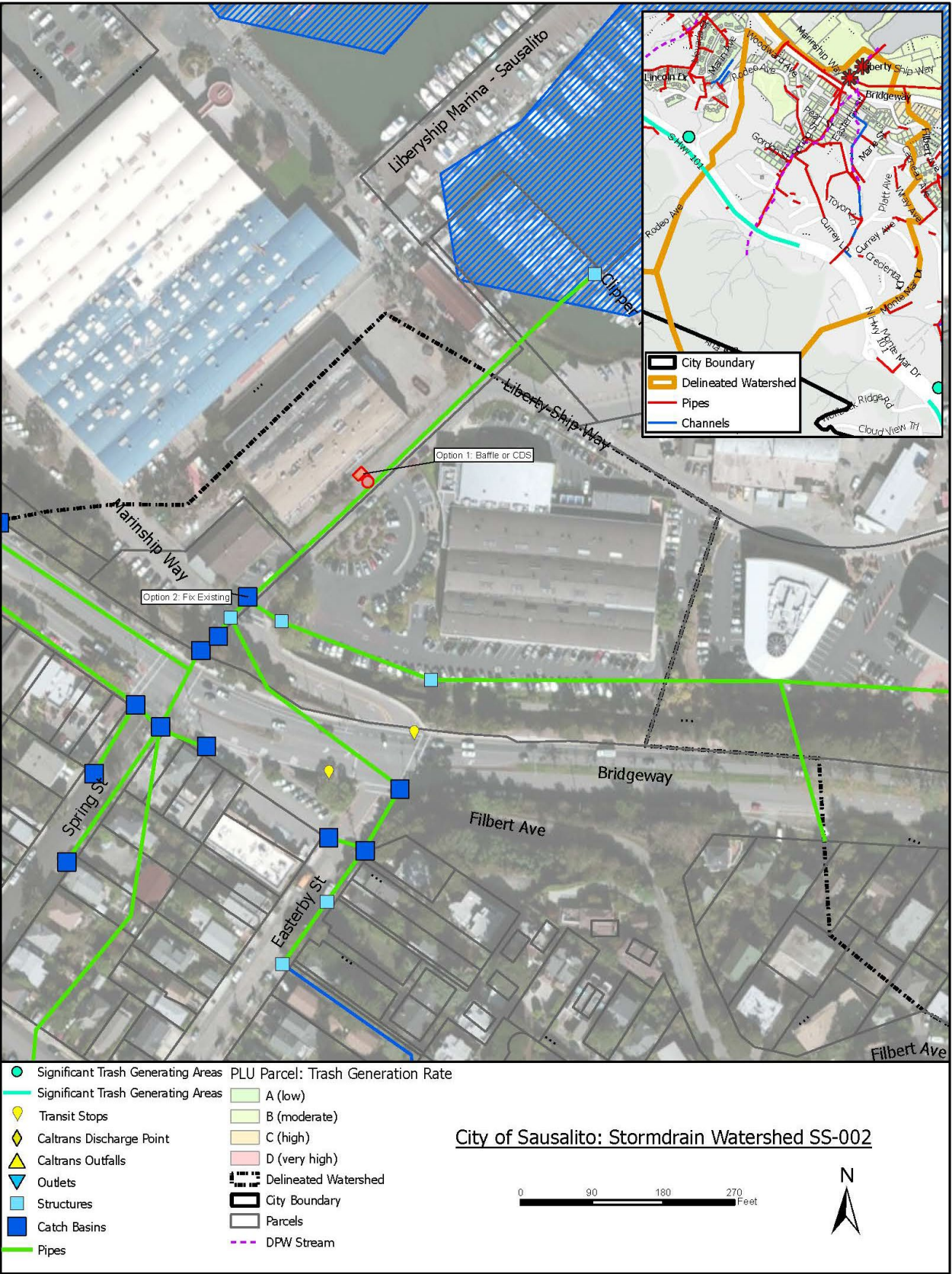


| Project ID: SS-002-Alt1                                                                                                                                                                                                                                                                                                                                  |                           |     |      |           |              |            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-----|------|-----------|--------------|------------|
| Project Name: Marinship Way                                                                                                                                                                                                                                                                                                                              |                           |     |      |           |              |            |
| Project Location: City of Sausalito (37.8623, -122.4944)<br>(Project drains Caltrans right-of-way area as well)                                                                                                                                                                                                                                          |                           |     |      |           |              |            |
| Site/Device Information                                                                                                                                                                                                                                                                                                                                  |                           |     |      |           |              |            |
| Proposed Device Type: Hydrodynamic Separator (HDS) or Baffle Box (See page 2 for Alternatives)                                                                                                                                                                                                                                                           |                           |     |      |           |              |            |
| Project Description: Construction of an underground hydrodynamic separator (HDS) or baffle box unit on Liberty Ship Way downstream from existing screen location that appears to be in a location that is not suitable for the hydraulics as the incoming pipe flow velocities are very high.                                                            |                           |     |      |           |              |            |
| Site Specific Considerations: <ul style="list-style-type: none"><li>Creek upstream of project site location</li><li>Private property ownership along Liberty Ship Way</li><li>Constant groundwater flow</li></ul>                                                                                                                                        |                           |     |      |           |              |            |
| O&M Summary: Requires a vacuum-assisted truck and traffic control/lane closure                                                                                                                                                                                                                                                                           |                           |     |      |           |              |            |
| Comments/Notes: Project will likely be considered a CEQA Categorical Exemption. Project will require coordination with the SF Bay Regional Water Board (Water Board), which oversees water quality certification programs for creeks/channels in the Bay Area. Project is dependent on Water Board's permission to treat stormwater downstream of creek. |                           |     |      |           |              |            |
| Drainage Area and Trash Reduction Information                                                                                                                                                                                                                                                                                                            |                           |     |      |           |              |            |
| Total Drainage Area: 232 acres <ul style="list-style-type: none"><li>Sausalito – 196 acres</li><li>Caltrans Right-of-Way – 36 acres</li></ul>                                                                                                                                                                                                            |                           |     |      |           |              |            |
| Priority Land Use (PLU) Areas (acres): 26 acres                                                                                                                                                                                                                                                                                                          |                           |     |      |           |              |            |
| Public Agency                                                                                                                                                                                                                                                                                                                                            | Trash Generation Category |     |      |           | Non-PLU Area | Total Area |
|                                                                                                                                                                                                                                                                                                                                                          | Low                       | Mod | High | Very High |              |            |
| Marin County                                                                                                                                                                                                                                                                                                                                             | 6                         | 19  | 1    | 0         | 206          | 232        |
| Caltrans                                                                                                                                                                                                                                                                                                                                                 | --                        | --  | --   | --        | --           | 36         |
| 1-yr/1-hr Peak Flow (modeled): 49 cfs                                                                                                                                                                                                                                                                                                                    |                           |     |      |           |              |            |

Proposed Project Location

Preliminary Cost Estimate

|                                           |                    |
|-------------------------------------------|--------------------|
| Design/Permitting                         | \$156,000          |
| Construction                              | \$593,000          |
| Construction Contingency                  | \$119,000          |
| <b>Total Capital</b>                      | <b>\$1,014,000</b> |
| <b>Annual Operation &amp; Maintenance</b> | <b>\$23,400/yr</b> |
| <b>Construction Cost per Acre of PLU</b>  | <b>\$32,800/ac</b> |







| Alternative Design Options |                                 |                                                                                                                                               |                                                                                                                                                                                |                                                                   |                                                                                                                                                                                                                                                                                                                                        |                                   |              |                                                                                                                          |
|----------------------------|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|--------------|--------------------------------------------------------------------------------------------------------------------------|
| Alternative Site ID        | Proposed Device Type            | Project Description                                                                                                                           | Site Specific Considerations                                                                                                                                                   | O&M Summary                                                       | Comments/Notes                                                                                                                                                                                                                                                                                                                         | Preliminary Cost Estimate         |              | Photos                                                                                                                   |
| SS-002-Alt2                | Modification of Existing Screen | Modification of an existing screen in the underground stormwater structure located at the intersection of Marinship Way and Liberty Ship Way. | <ul style="list-style-type: none"><li>Creek upstream of project site location</li><li>Large and constant flow in the structure at Marinship Way and Liberty Ship Way</li></ul> | Requires a vacuum-assisted truck and traffic control/lane closure | City of Sausalito is interested in fixing the existing grate. Project will require coordination with the SF Bay Regional Water Board (Water Board), which oversees water quality certification programs for creeks/channels in the Bay Area. Project is dependent on Water Board's permission to treat stormwater downstream of creek. | Design/Permitting                 | N/A          | <div>Proposed Project Location</div>  |
|                            |                                 |                                                                                                                                               |                                                                                                                                                                                |                                                                   |                                                                                                                                                                                                                                                                                                                                        | Construction                      | N/A          |                                                                                                                          |
|                            |                                 |                                                                                                                                               |                                                                                                                                                                                |                                                                   |                                                                                                                                                                                                                                                                                                                                        | Construction Contingency          | N/A          |                                                                                                                          |
|                            |                                 |                                                                                                                                               |                                                                                                                                                                                |                                                                   |                                                                                                                                                                                                                                                                                                                                        | Total Capital                     | N/A          |                                                                                                                          |
|                            |                                 |                                                                                                                                               |                                                                                                                                                                                |                                                                   |                                                                                                                                                                                                                                                                                                                                        | Annual Operation & Maintenance    | N/A          |                                                                                                                          |
|                            |                                 |                                                                                                                                               |                                                                                                                                                                                |                                                                   |                                                                                                                                                                                                                                                                                                                                        | Construction Cost per Acre of PLU | N/A          |                                                                                                                          |
| SS-002-Alt3                | Small Devices                   | This proposed alternative would entail the construction of approximately 54 inlet screens throughout the drainage area of this project.       | N/A                                                                                                                                                                            | Requires maintenance at many different locations.                 | Construction Contingency cost included in the overall Construction cost.                                                                                                                                                                                                                                                               | Design/Permitting                 | \$27,000     | N/A                                                                                                                      |
|                            |                                 |                                                                                                                                               |                                                                                                                                                                                |                                                                   |                                                                                                                                                                                                                                                                                                                                        | Construction                      | \$135,000    |                                                                                                                          |
|                            |                                 |                                                                                                                                               |                                                                                                                                                                                |                                                                   |                                                                                                                                                                                                                                                                                                                                        | Total Capital                     | \$162,000    |                                                                                                                          |
|                            |                                 |                                                                                                                                               |                                                                                                                                                                                |                                                                   |                                                                                                                                                                                                                                                                                                                                        | Annual Operation & Maintenance    | \$389,000/yr |                                                                                                                          |
|                            |                                 |                                                                                                                                               |                                                                                                                                                                                |                                                                   |                                                                                                                                                                                                                                                                                                                                        | Construction Cost per Acre of PLU | \$6,200/ac   |                                                                                                                          |



CONCEPTUAL DESIGN - POTENTIAL FULL TRASH SYSTEM PROJECT



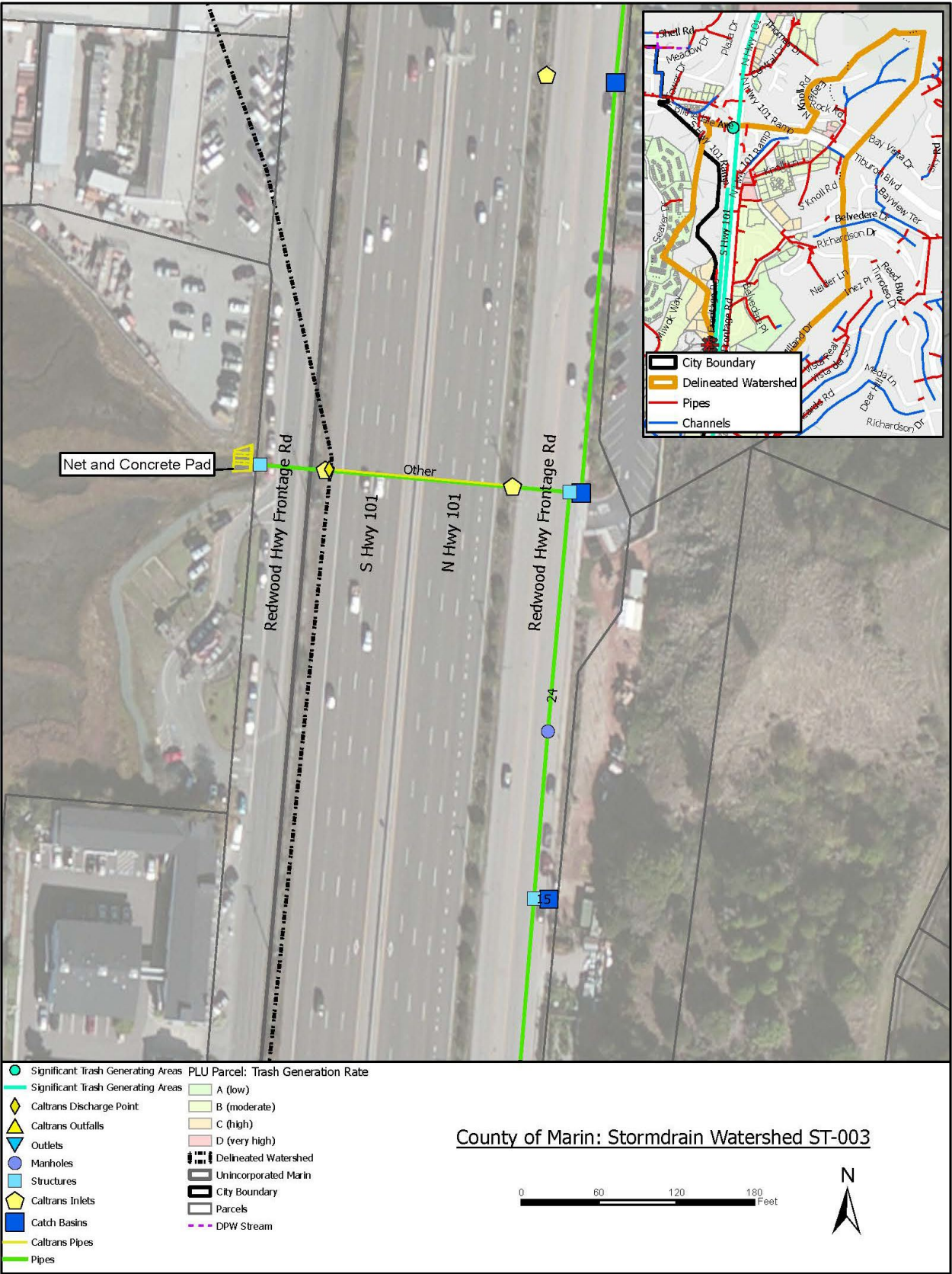
|                                                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Project ID:</b> ST-003-Alt1                                                                                                                       |
| <b>Project Name:</b> Redwood Highway                                                                                                                 |
| <b>Project Location:</b> Marin County ( <a href="#">37.8945</a> , <a href="#">-122.5168</a> )<br>(Project drains Caltrans right-of-way area as well) |

|                                                                                                                                                                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Site/Device Information</b>                                                                                                                                                                                                                                               |
| <b>Proposed Device Type:</b> Netting Device ( <a href="#">See page 2 for Alternatives</a> )                                                                                                                                                                                  |
| <b>Project Description:</b> Construction of an end-of-pipe netting device and concrete pad. The netting device would be attached to the headwall of the pipe that discharges into the pond located adjacent to Cable Car Coffee.                                             |
| <b>Site Specific Considerations:</b> <ul style="list-style-type: none"><li>Pond area is private property</li><li>Location is adjacent to multiple businesses and parking area</li><li>Near tidal water body</li><li>Cofferdam will be required during construction</li></ul> |
| <b>O&amp;M Summary:</b> May require drawdown if standing water is present when maintenance is planned, or needs to be conducted at low tide.                                                                                                                                 |
| <b>Comments/Notes:</b> Project will require coordination with private landowners who own the pond area. Area is tidal wetland.                                                                                                                                               |



| Drainage Area and Trash Reduction Information                                                                                                           |                           |     |      |           |              |            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-----|------|-----------|--------------|------------|
| <b>Total Drainage Area:</b> 185 acres <ul style="list-style-type: none"><li>Marin County – 155 acres</li><li>Caltrans Right-of-Way – 30 acres</li></ul> |                           |     |      |           |              |            |
| <b>Priority Land Use (PLU) Areas (acres):</b> 37 acres                                                                                                  |                           |     |      |           |              |            |
| Public Agency                                                                                                                                           | Trash Generation Category |     |      |           | Non-PLU Area | Total Area |
|                                                                                                                                                         | Low                       | Mod | High | Very High |              |            |
| Marin County                                                                                                                                            | 0                         | 30  | 5    | 2         | ?            | 155        |
| Caltrans                                                                                                                                                | --                        | --  | --   | --        | --           | 30         |
| <b>1-yr/1-hr Peak Flow (modeled):</b> 54 cfs                                                                                                            |                           |     |      |           |              |            |

| Preliminary Cost Estimate                 |                    |
|-------------------------------------------|--------------------|
| Design/Permitting                         | \$212,000          |
| Construction                              | \$403,000          |
| Construction Contingency                  | \$80,600           |
| <b>Total Capital</b>                      | <b>\$826,000</b>   |
| <b>Annual Operation &amp; Maintenance</b> | <b>\$27,300/yr</b> |
| <b>Construction Cost per Acre of PLU</b>  | <b>\$16,600/ac</b> |







| Alternative Design Options |                                            |                                                                                                                                          |                                     |                                                                   |                                                                                                                                     |                                   |                |                                                                                                                          |
|----------------------------|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|-------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|----------------|--------------------------------------------------------------------------------------------------------------------------|
| Alternative Site ID        | Proposed Device Type                       | Project Description                                                                                                                      | Site Specific Considerations        | O&M Summary                                                       | Comments/Notes                                                                                                                      | Preliminary Cost Estimate         |                | Photos                                                                                                                   |
| ST-003-Alt2                | Hydrodynamic Separator (HDS) or Baffle Box | Construction of an underground HDS or baffle box unit beneath Redwood Highway.                                                           | High traffic area, high groundwater | Requires a vacuum-assisted truck and traffic control/lane closure | Construction of the device would occupy much of the street and require complete street closure during construction and maintenance. | Design/Permitting                 | \$171,500      | <div>Proposed Project Location</div>  |
|                            |                                            |                                                                                                                                          |                                     |                                                                   |                                                                                                                                     | Construction                      | \$810,000      |                                                                                                                          |
|                            |                                            |                                                                                                                                          |                                     |                                                                   |                                                                                                                                     | Construction Contingency          | \$129,000      |                                                                                                                          |
|                            |                                            |                                                                                                                                          |                                     |                                                                   |                                                                                                                                     | Total Capital                     | \$982,000      |                                                                                                                          |
|                            |                                            |                                                                                                                                          |                                     |                                                                   |                                                                                                                                     | Annual Operation & Maintenance    | \$23,400/yr    |                                                                                                                          |
|                            |                                            |                                                                                                                                          |                                     |                                                                   |                                                                                                                                     | Construction Cost per Acre of PLU | \$21,900/ac    |                                                                                                                          |
| ST-003-Alt3                | Small Devices                              | This proposed alternative would entail the construction of approximately 162 inlet screens throughout the drainage area of this project. | N/A                                 | Requires maintenance at many different locations.                 | Construction Contingency cost included in the overall Construction cost.                                                            | Design/Permitting                 | \$81,000       | N/A                                                                                                                      |
|                            |                                            |                                                                                                                                          |                                     |                                                                   |                                                                                                                                     | Construction                      | \$405,000      |                                                                                                                          |
|                            |                                            |                                                                                                                                          |                                     |                                                                   |                                                                                                                                     | Total Capital                     | \$486,000      |                                                                                                                          |
|                            |                                            |                                                                                                                                          |                                     |                                                                   |                                                                                                                                     | Annual Operation & Maintenance    | \$1,170,000/yr |                                                                                                                          |
|                            |                                            |                                                                                                                                          |                                     |                                                                   |                                                                                                                                     | Construction Cost per Acre of PLU | \$9,500/ac     |                                                                                                                          |



CONCEPTUAL DESIGN - POTENTIAL FULL TRASH SYSTEM PROJECT



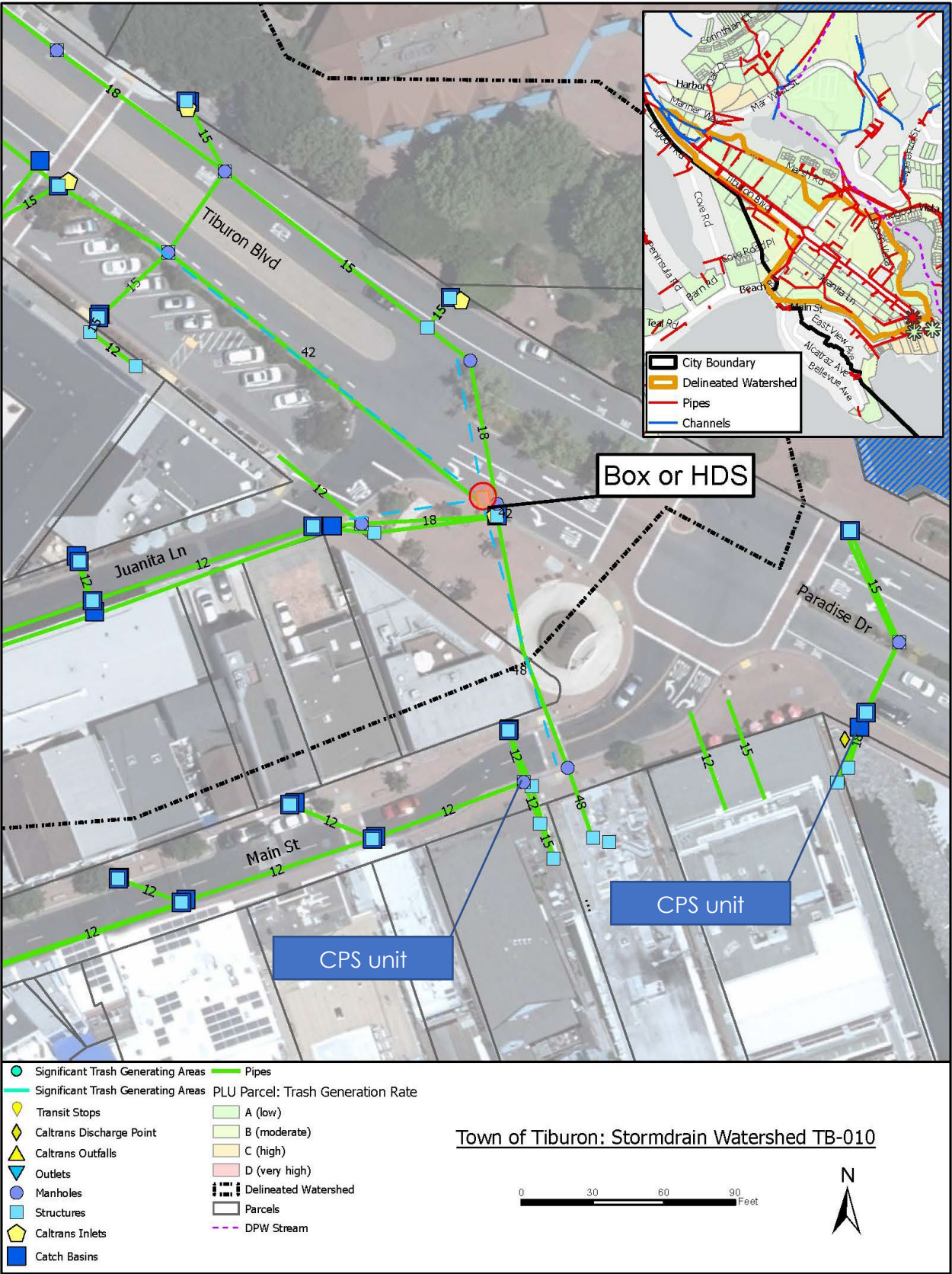
|                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Project ID:</b> TB-010-Alt1                                                                                                         |
| <b>Project Name:</b> Paradise Drive                                                                                                    |
| <b>Project Location:</b> Town of Tiburon ( <a href="#">37.8736, -122.4561</a> )<br>(Project drains Caltrans right-of-way area as well) |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Site/Device Information</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Proposed Device Type:</b> Hydrodynamic Separator (HDS) or Baffle Box and Connector Pipe Screens (CPSs) on Main Street (See page 2 for Alternatives)                                                                                                                                                                                                                                                                                                                                         |
| <b>Project Description:</b> Construction of an underground hydrodynamic separator (HDS) or baffle box unit. The large device would be installed within an existing 48-inch pipe located beneath the turn lane on Paradise Drive. The device would be upstream of a flap gate, which should be replaced as part of this project. Replacement of the flap gate is not included in the cost at this time. Additional small devices (CPS devices) would be installed within inlets on Main Street. |
| <b>Site Specific Considerations:</b> <ul style="list-style-type: none"><li>Location is known to have capacity issues related to high tides</li><li>Flap gate downstream of project location will require replacement</li><li>Site will require coordination with Caltrans</li></ul>                                                                                                                                                                                                            |
| <b>O&amp;M Summary:</b> Requires a vacuum-assisted truck and traffic control/lane closure                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Comments/Notes:</b> Project will likely be considered a CEQA Categorical Exemption.                                                                                                                                                                                                                                                                                                                                                                                                         |

| Drainage Area and Trash Reduction Information                                                                                                   |                           |     |      |           |              |            |
|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-----|------|-----------|--------------|------------|
| <b>Total Drainage Area:</b> 26 acres <ul style="list-style-type: none"><li>Tiburon – 21 acres</li><li>Caltrans Right-of-Way – 5 acres</li></ul> |                           |     |      |           |              |            |
| <b>Priority Land Use (PLU) Areas (acres):</b> 18 acres                                                                                          |                           |     |      |           |              |            |
| Public Agency                                                                                                                                   | Trash Generation Category |     |      |           | Non-PLU Area | Total Area |
|                                                                                                                                                 | Low                       | Mod | High | Very High |              |            |
| Tiburon                                                                                                                                         | 8                         | 10  | 0    | 0         | 3            | 21         |
| Caltrans                                                                                                                                        | --                        | --  | --   | --        | --           | 5          |
| <b>1-yr/1-hr Peak Flow (modeled):</b> 11 cfs                                                                                                    |                           |     |      |           |              |            |



| Preliminary Cost Estimate                 |                    |
|-------------------------------------------|--------------------|
| Design/Permitting                         | \$56,000           |
| Construction                              | \$134,000          |
| Construction Contingency                  | \$27,000           |
| <b>Total Capital</b>                      | <b>\$250,000</b>   |
| <b>Annual Operation &amp; Maintenance</b> | <b>\$23,400/yr</b> |
| <b>Construction Cost per Acre of PLU</b>  | <b>\$10,800/ac</b> |





CONCEPTUAL DESIGN - POTENTIAL FULL TRASH SYSTEM PROJECT



| Alternative Design Options |                      |                                                                                                                                          |                              |                                                   |                                                                          |                                   |                |        |
|----------------------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|---------------------------------------------------|--------------------------------------------------------------------------|-----------------------------------|----------------|--------|
| Alternative Site ID        | Proposed Device Type | Project Description                                                                                                                      | Site Specific Considerations | O&M Summary                                       | Comments/Notes                                                           | Preliminary Cost Estimate         |                | Photos |
| TB-010-Alt2                | Small Devices        | This proposed alternative would entail the construction of approximately 200 inlet screens throughout the drainage area of this project. | N/A                          | Requires maintenance at many different locations. | Construction Contingency cost included in the overall Construction cost. | Design/ Permitting                | \$100,000      | N/A    |
|                            |                      |                                                                                                                                          |                              |                                                   |                                                                          | Construction                      | \$500,000      |        |
|                            |                      |                                                                                                                                          |                              |                                                   |                                                                          | Total Capital                     | \$600,000      |        |
|                            |                      |                                                                                                                                          |                              |                                                   |                                                                          | Annual Operation & Maintenance    | \$1,440,000/yr |        |
|                            |                      |                                                                                                                                          |                              |                                                   |                                                                          | Construction Cost per Acre of PLU | \$33,400/ac    |        |

CONCEPTUAL DESIGN - POTENTIAL FULL TRASH SYSTEM PROJECT



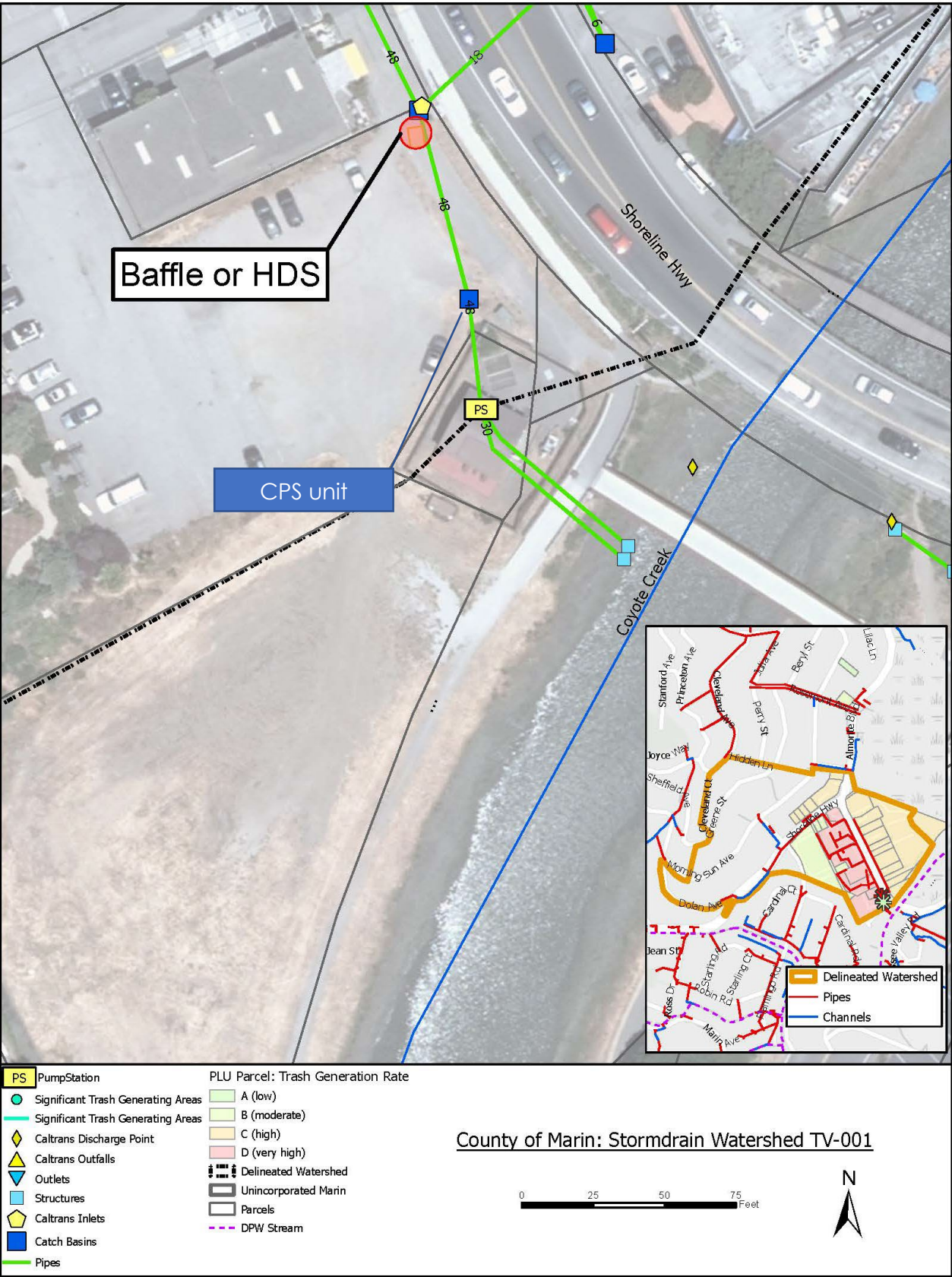
| Project ID: TV-001-A1#1                                                                                                                                                                                                                                                                                 |                           |     |      |           |              |            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-----|------|-----------|--------------|------------|
| Project Name: Shoreline PS                                                                                                                                                                                                                                                                              |                           |     |      |           |              |            |
| Project Location: Marin County ( <a href="#">37.8799, -122.5240</a> )<br>(Project drains Caltrans right-of-way area as well)                                                                                                                                                                            |                           |     |      |           |              |            |
| Site/Device Information                                                                                                                                                                                                                                                                                 |                           |     |      |           |              |            |
| Proposed Device Type: Hydrodynamic Separator (HDS) or Baffle Box and Connector Pipe Screen (CPS) (See page 2 for Alternatives)                                                                                                                                                                          |                           |     |      |           |              |            |
| Project Description: Construction of an underground hydrodynamic separator (HDS) or baffle box unit. The large device would be installed within an existing 48-inch pipe located beneath the bike pathway along Shoreline Highway. One additional small device (CPS device) would be installed as well. |                           |     |      |           |              |            |
| Site Specific Considerations: <ul style="list-style-type: none"><li>Location is close to Caltrans ROW for Shoreline Highway</li><li>Location is on a bike path</li><li>Construction easement will be required for constructing in the pathway</li></ul>                                                 |                           |     |      |           |              |            |
| O&M Summary: Requires a vacuum-assisted truck and closure of bike pathway.                                                                                                                                                                                                                              |                           |     |      |           |              |            |
| Comments/Notes: Project will likely be considered a CEQA Categorical Exemption. Preliminary cost estimate does not include the one CPS device.                                                                                                                                                          |                           |     |      |           |              |            |
| Drainage Area and Trash Reduction Information                                                                                                                                                                                                                                                           |                           |     |      |           |              |            |
| Total Drainage Area: 42 acres <ul style="list-style-type: none"><li>Marin County – 38 acres</li><li>Caltrans Right-of-Way – 4 acres</li></ul>                                                                                                                                                           |                           |     |      |           |              |            |
| Priority Land Use (PLU) Areas (acres): 17 acres                                                                                                                                                                                                                                                         |                           |     |      |           |              |            |
| Public Agency                                                                                                                                                                                                                                                                                           | Trash Generation Category |     |      |           | Non-PLU Area | Total Area |
|                                                                                                                                                                                                                                                                                                         | Low                       | Mod | High | Very High |              |            |
| Marin County                                                                                                                                                                                                                                                                                            | 0                         | 3   | 9    | 5         | 21           | 38         |
| Caltrans                                                                                                                                                                                                                                                                                                | --                        | --  | --   | --        | --           | 4          |
| 1-yr/1-hr Peak Flow (modeled): 15 cfs                                                                                                                                                                                                                                                                   |                           |     |      |           |              |            |

Proposed Project Location

Proposed Project Location

Preliminary Cost Estimate

|                                           |                    |
|-------------------------------------------|--------------------|
| Design/Permitting                         | \$65,000           |
| Construction                              | \$174,000          |
| Construction Contingency                  | \$34,900           |
| <b>Total Capital</b>                      | <b>\$319,000</b>   |
| <b>Annual Operation &amp; Maintenance</b> | <b>\$23,400/yr</b> |
| <b>Construction Cost per Acre of PLU</b>  | <b>\$14,800/ac</b> |







| Alternative Design Options |                      |                                                                                                                                          |                              |                                                   |                                                                          |                                   |              |        |
|----------------------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|---------------------------------------------------|--------------------------------------------------------------------------|-----------------------------------|--------------|--------|
| Alternative Site ID        | Proposed Device Type | Project Description                                                                                                                      | Site Specific Considerations | O&M Summary                                       | Comments/Notes                                                           | Preliminary Cost Estimate         |              | Photos |
| TV-001-Alt2                | Small Devices        | This proposed alternative would entail the construction of approximately 115 inlet screens throughout the drainage area of this project. | N/A                          | Requires maintenance at many different locations. | Construction Contingency cost included in the overall Construction cost. | Design/ Permitting                | \$57,500     | N/Asss |
|                            |                      |                                                                                                                                          |                              |                                                   |                                                                          | Construction                      | \$288,000    |        |
|                            |                      |                                                                                                                                          |                              |                                                   |                                                                          | Total Capital                     | \$345,000    |        |
|                            |                      |                                                                                                                                          |                              |                                                   |                                                                          | Annual Operation & Maintenance    | \$828,000/yr |        |
|                            |                      |                                                                                                                                          |                              |                                                   |                                                                          | Construction Cost per Acre of PLU | \$20,000/ac  |        |

## **ATTACHMENT 2**

RWQCB Approved Device List



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## State Water Resources Control Board

### **CERTIFIED FULL CAPTURE SYSTEM LIST OF TRASH TREATMENT CONTROL DEVICES (Updated September 2021)**

#### **Trash Provisions**

In accordance with the Trash Provisions,<sup>1</sup> all trash treatment control devices (Devices) installed after December 2, 2015 shall meet the Full Capture System definition<sup>2</sup> and be certified by the State Water Resources Control Board (State Water Board) Executive Director, or designee, prior to installation. The Devices included on this list are either: 1) new Device applications certified by the State Water Board after adoption of the Trash Provisions, or 2) grandfathered device fact sheets listed by the San Francisco Regional Water Board prior to the adoption of the Trash Provisions.

The Trash Provisions require that only Certified Full Capture Devices may be installed, and that the Devices be designed according to the following criteria:

1. Appropriately sized to treat not less than the peak flowrate resulting from a one--year, one-hour storm event (design storm) or at least the same peak flows from the corresponding storm drain;
2. Do not bypass trash below the design storm under maximum operational loading conditions; and
3. Trap all particles that are 5 mm or greater up to the design flow<sup>3</sup> or at least the same peak flows from the corresponding storm drain; and do not have a

---

<sup>1</sup> Amendment to the Water Quality Control Plan for Ocean Waters of California to Control Trash and Part 1 Trash Provisions of the Water Quality Control Plan for Inland Surface Waters, Enclosed Bays, And Estuaries of California adopted by the State Water Board.

<sup>2</sup> A Full Capture System is a treatment control, or series of treatment controls, including but not limited to, a multi-benefit project or a low-impact development control that traps all particles that are 5 mm or greater, and has a design treatment capacity that is either: of not less than the peak flow rate, Q, resulting from a one-year, one-hour storm in the subdrainage area, or b) appropriately sized to, and designed to carry at least the same flows as, the corresponding storm drain.

<sup>3</sup> The region specific one-year, one-hour storm (or design flow) may be obtained from the [National Oceanic and Atmospheric Precipitation Estimates](https://www.weather.gov/media/owp/oh/hdsc/docs/Atlas14_Volume6.pdf) ([https://www.weather.gov/media/owp/oh/hdsc/docs/Atlas14\\_Volume6.pdf](https://www.weather.gov/media/owp/oh/hdsc/docs/Atlas14_Volume6.pdf))

## CERTIFIED FULL CAPTURE SYSTEM LIST OF TRASH TREATMENT CONTROL DEVICES

diversion structure present upstream such that a portion of the peak flow is not treated to trap all particles 5-millimeter or greater.

### Vector Control Accessibility

According to the California Health and Safety Code<sup>4</sup>, California landowners are legally responsible to abate (eliminate the source of) a public nuisance arising from their property, including mosquitoes. Mosquito vector control districts have substantial authority to access public and private property, inspect known or suspected sources of mosquitoes, abate mosquito sources, and charge the landowner for work performed and/or charge fees if a landowner is unwilling or unable to address a mosquito source arising from their property.

Depending on its design, certain Devices may impede the mosquito vector control district's ability to (1) visually inspect the Device and/or storm vault for mosquito breeding, and (2) apply the appropriate chemical treatment. Moreover, some devices may create a habitat for mosquitoes. Prior to installation of any certified Device, the local mosquito vector control district should be contacted to ensure the installation conforms to the district's visual inspection, treatment, and vector breeding minimizing guidelines. The Mosquito Vector Control Association of California may also be contacted via email at [Trashtreatment@mvcac.org](mailto:Trashtreatment@mvcac.org).

### New Device Application Certification or Fact Sheet Update

To apply for certification of a new Device, or to update a grandfathered Device fact sheet, the Device owner shall submit an application/fact sheet in accordance with the *Trash Treatment Control Device Certification and Fact Sheet Update Requirements*. Upon determining that a Device application is complete and meets the definition of a trash full capture system and is approved by the Mosquito Vector Control Association of California, the Executive Director will place the Device on the State Water Board's *Certified Full Capture System List of Trash Treatment Control Devices*.

The Trash Treatment Control Device Certification and Fact Sheet Update Requirements is found on the [Trash Implementation Program webpage](https://www.waterboards.ca.gov/water_issues/programs/stormwater/trash_implementation.html) ([https://www.waterboards.ca.gov/water\\_issues/programs/stormwater/trash\\_implementation.html](https://www.waterboards.ca.gov/water_issues/programs/stormwater/trash_implementation.html)).

Certification of any Device does not constitute an endorsement by the State Water Board. The Executive Director reserves the right to de-certify and remove any Device from this list that does not satisfy the requirements of the Trash Provisions, such as but not limited to when a Device is discontinued, found to not trap trash in accordance with the Trash Provisions, or no longer has Mosquito Vector Control Association of California approval.

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<sup>4</sup> Health & Safety Code sections 2001- 4(d); 2002; 2060 (b) and Health & Safety Code sections 2060-2067, 100170, and 100175.



## **CERTIFIED FULL CAPTURE SYSTEM LIST OF TRASH TREATMENT CONTROL DEVICES**

Device applications and fact sheets may be obtained at:

<https://www.casqa.org/resources/trash/certified-full-capture-system-trash-treatment-control-devices>.

Questions regarding certification should be directed to Leo Cosentini at (916) 341-5524 or email address ([leo.cosentini@waterboards.ca.gov](mailto:leo.cosentini@waterboards.ca.gov)).

**CERTIFIED FULL CAPTURE SYSTEM LIST OF  
TRASH TREATMENT CONTROL DEVICES**

**TABLE 1 - CATCH BASIN INSERTS AND OTHER DEVICES**

| <b>Owner</b>                                              | <b>Full Capture System Trash<br/>Device Brand Name</b> | <b>Date Application<br/>Certified or Fact<br/>Sheet Updated</b> | <b>Date Vector<br/>Control<br/>Accessibility<br/>Verified</b> |
|-----------------------------------------------------------|--------------------------------------------------------|-----------------------------------------------------------------|---------------------------------------------------------------|
| <b>AbTech, Industries<sup>5</sup></b>                     | <b>Ultra Urban Filter Curb<br/>Opening and Drop-In</b> | <b>Application 25<br/>05/01/20</b>                              | <b>04/8/20</b>                                                |
| <b>Advanced Drainage Systems, Inc.<sup>6</sup></b>        | <b>FLEXSTORM PURE<br/>Full Trash Capture Inserts</b>   | <b>Application 3<br/>03/15/18<br/>Updated 04/21/21</b>          | <b>03/30/21</b>                                               |
| <b>Advanced Drainage Systems, Inc.</b>                    | <b>FLEXSTORM<br/>Connector Pipe Screen</b>             | <b>ADS-1<br/>Updated 06/08/21</b>                               | <b>03/30/21</b>                                               |
| <b>Bio Clean Environmental Services, Inc.<sup>7</sup></b> | <b>Curb Inlet and Grate<br/>Inlet Filters</b>          | <b>Application 4<br/>03/15/18<br/>Updated 02/12/21</b>          | <b>01/27/21</b>                                               |
| <b>Bio Clean Environmental Services, Inc.</b>             | <b>Modular Connector Pipe<br/>Trash Screen</b>         | <b>BC-3<br/>Updated 04/30/20</b>                                | <b>03/10/20</b>                                               |
| <b>BrightWater<sup>8</sup></b>                            | <b>Connector Pipe Screen</b>                           | <b>Application 29<br/>12/28/20</b>                              | <b>11/19/20</b>                                               |

<sup>5</sup> AbTech Industries website: <https://www.abtechindustries.com/>

<sup>6</sup> Advanced Drainage Systems, Inc. website: <http://www.inletfilters.com/>

<sup>7</sup> Bio Clean Environmental Services, Inc. website: <http://www.biocleanenvironmental.com/products/>

<sup>8</sup> BrightWater website: [www.wearebrightwater.com/](http://www.wearebrightwater.com/)



**CERTIFIED FULL CAPTURE SYSTEM LIST OF  
TRASH TREATMENT CONTROL DEVICES**

| <b>Owner</b>                                                                | <b>Full Capture System Trash Device Brand Name</b>                                        | <b>Date Application Certified or Fact Sheet Updated</b> | <b>Date Vector Control Accessibility Verified</b> |
|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|---------------------------------------------------------|---------------------------------------------------|
| <b>BrightWater</b>                                                          | <b>Curb Inlet Filter</b>                                                                  | <b>Application 26<br/>06/30/20</b>                      | <b>04/17/20</b>                                   |
| <b>Ecology Control Industries<sup>9</sup></b>                               | <b>Debris Dam - Catch Basin Insert for Curb Inlet Design</b>                              | <b>ECI-1<br/>Updated 06/17/20</b>                       | <b>04/29/20</b>                                   |
| <b>Enviropod International: A Stormwater 360 Group Company<sup>10</sup></b> | <b>Enviropod® LittaTrap™ Full Capture</b>                                                 | <b>Application 27<br/>10/15/20</b>                      | <b>07/20/20</b>                                   |
| <b>Filtrexx Sustainable Technologies<sup>11</sup></b>                       | <b>StormExx® Clean</b>                                                                    | <b>Application 16<br/>08/10/18<br/>Updated 11/25/19</b> | <b>12/06/19</b>                                   |
| <b>Frog Creek Partners, LLC<sup>12</sup></b>                                | <b>Gutter Bin® Channel Filtration System &amp; Mundus Bag® Water Filter</b>               | <b>Application 22<br/>06/26/19</b>                      | <b>04/19/19</b>                                   |
| <b>Frog Creek Partners, LLC</b>                                             | <b>Gutter Bin® Eco Drop Inlet Filter (DIF &amp; DIF-C) &amp; Mundus Bag® Water Filter</b> | <b>Application 24<br/>02/18/20</b>                      | <b>12/06/19</b>                                   |

<sup>9</sup> Ecology Control Industries website: <http://www.ecologycontrol.com/>

<sup>10</sup> Enviropod International: A Stormwater 360 Group Company website: <https://www.enviropod.com/products/enviropod-littatrap-full-capture/>

<sup>11</sup> Filtrexx Sustainable Technologies website: <https://www.filtrexx.com/en/products/stormexx/>

<sup>12</sup> Frog Creek Partners, LLC website: <https://frogcreek.partners/>

**CERTIFIED FULL CAPTURE SYSTEM LIST OF  
TRASH TREATMENT CONTROL DEVICES**

| <b>Owner</b>                                  | <b>Full Capture System Trash Device Brand Name</b>                      | <b>Date Application Certified or Fact Sheet Updated</b> | <b>Date Vector Control Accessibility Verified</b> |
|-----------------------------------------------|-------------------------------------------------------------------------|---------------------------------------------------------|---------------------------------------------------|
| <b>Frog Creek Partners, LLC</b>               | <b>Gutter Bin® Eco Curb Inlet Filter &amp; Mundus Bag® Water Filter</b> | <b>Application 23<br/>02/18/20</b>                      | <b>10/11/19</b>                                   |
| <b>G2 Construction, Inc.<sup>13</sup></b>     | <b>G2 CPS-Mod™ and Removable CPS-Mod™ Screen</b>                        | <b>Application 18<br/>06/26/19</b>                      | <b>03/15/19</b>                                   |
| <b>G2 Construction, Inc.</b>                  | <b>G2 Grated Inlet Trash Screen</b>                                     | <b>Application 19<br/>06/26/19</b>                      | <b>04/10/19</b>                                   |
| <b>Inventive Resources, Inc.<sup>14</sup></b> | <b>Water Decontaminator</b>                                             | <b>Application 2<br/>03/15/18<br/>Updated 02/05/21</b>  | <b>04/20/20</b>                                   |
| <b>Oldcastle Infrastructure<sup>15</sup></b>  | <b>FloGard® Curb Inlet Basket</b>                                       | <b>OI-1<br/>Updated 06/09/21</b>                        | <b>06/09/21</b>                                   |
| <b>Oldcastle Infrastructure</b>               | <b>FloGard® Grate Inlet Basket</b>                                      | <b>OI-2<br/>Updated 06/09/21</b>                        | <b>06/09/21</b>                                   |

<sup>13</sup> G2 Construction, Inc website: <http://www.g2construction.com/products/>

<sup>14</sup> Inventive Resources, Inc. website: <http://www.IRIproducts.com/>

<sup>15</sup> Oldcastle Infrastructure website: <https://oldcastleinfrastructure.com/brands/>



**CERTIFIED FULL CAPTURE SYSTEM LIST OF  
TRASH TREATMENT CONTROL DEVICES**

| <b>Owner</b>                                                | <b>Full Capture System Trash Device Brand Name</b>                                           | <b>Date Application Certified or Fact Sheet Updated</b> | <b>Date Vector Control Accessibility Verified</b> |
|-------------------------------------------------------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------|---------------------------------------------------|
| <b>Oldcastle Infrastructure</b>                             | <b>FloGard® Outlet Trash Screen</b>                                                          | <b>OI-3<br/>Updated 11/29/19</b>                        | <b>12/06/19</b>                                   |
| <b>Revel Environmental Manufacturing, Inc.<sup>16</sup></b> | <b>Triton™ Bioflex Inlet Trash Guard Catchbasin Polyester Fiber Mesh Trash Filter Insert</b> | <b>REM-1<br/>Updated 09/10/21</b>                       | <b>09/07/21</b>                                   |
| <b>Revel Environmental Manufacturing, Inc.</b>              | <b>Triton™ Crescent Pipe Screen</b>                                                          | <b>Application 12<br/>07/10/18</b>                      | <b>03/15/19</b>                                   |
| <b>Revel Environmental Manufacturing, Inc.</b>              | <b>Triton Perf-Full Trash Capture Insert</b>                                                 | <b>Application 13<br/>07/10/18</b>                      | <b>03/15/19</b>                                   |
| <b>Safe Drain Stormwater Holdings Inc.<sup>17</sup></b>     | <b>Storm Vector Guard</b>                                                                    | <b>Application 30<br/>02/11/21</b>                      | <b>12/17/20</b>                                   |
| <b>Stormtek<sup>18</sup></b>                                | <b>Stormtek ST3 &amp; STEG Catchbasin Connector Pipe</b>                                     | <b>AS-1, A1S-2<br/>Updated 08/12/21</b>                 | <b>08/04/21</b>                                   |

<sup>16</sup> Revel Environmental Manufacturing, Inc. website: <http://www.remfilters.com/>

<sup>17</sup> Safe Drain Stormwater Holdings, Inc. website: <http://www.safedrainusa.com/>

<sup>18</sup> Stormtek website: <https://swimsclean.com/stormtek/>

**CERTIFIED FULL CAPTURE SYSTEM LIST OF  
TRASH TREATMENT CONTROL DEVICES**

| Owner                                 | Full Capture System Trash Device Brand Name | Date Application Certified or Fact Sheet Updated | Date Vector Control Accessibility Verified |
|---------------------------------------|---------------------------------------------|--------------------------------------------------|--------------------------------------------|
| United Stormwater, Inc. <sup>19</sup> | Connector Pipe Trash Screen                 | USW-1<br>Updated 08/06/21                        | 08/05/21                                   |

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<sup>19</sup> United Stormwater, Inc. website: <http://www.unitedstormwater.com/>



**CERTIFIED FULL CAPTURE SYSTEM LIST OF  
TRASH TREATMENT CONTROL DEVICES**

**TABLE 2 - HIGH FLOW CAPACITY TRASH DEVICES**

| <b>Owner</b>                                                                     | <b>Full Capture System Trash<br/>Device Brand Name</b> | <b>Date Application<br/>Certified or Fact<br/>Sheet Updated</b> | <b>Date Vector<br/>Control<br/>Accessibility<br/>Verified</b> |
|----------------------------------------------------------------------------------|--------------------------------------------------------|-----------------------------------------------------------------|---------------------------------------------------------------|
| <b>AquaShield, Inc.<sup>20</sup></b>                                             | <b>Aqua-Swirl® Stormwater<br/>Treatment System</b>     | <b>Application 1<br/>08/04/17<br/>Updated 11/06/20</b>          | <b>12/03/20</b>                                               |
| <b>BaySaver Technologies LLC/Advanced<br/>Drainage Systems Inc.<sup>21</sup></b> | <b>Barracuda Hydrodynamic<br/>Separator</b>            | <b>Application 21<br/>06/26/19<br/>Updated 05/21/21</b>         | <b>03/15/19</b>                                               |
| <b>Bio Clean Environmental Services, Inc.</b>                                    | <b>Debris Separating Baffle<br/>Box</b>                | <b>Application 6<br/>03/15/18</b>                               | <b>07/28/20</b>                                               |
| <b>Bio Clean Environmental Services, Inc.</b>                                    | <b>BioClean<br/>Deflective Screening<br/>Device</b>    | <b>Application 20<br/>06/26/19</b>                              | <b>07/28/20</b>                                               |
| <b>Bio Clean Environmental Services, Inc.</b>                                    | <b>Modular Wetland System®</b>                         | <b>Application 15<br/>07/10/18</b>                              | <b>03/15/19</b>                                               |

<sup>20</sup> AquaShield, Inc. website: <http://www.aquashieldinc.com/-aqua-swirl.html>

<sup>21</sup> BaySaver Technologies LLC/Advanced Drainage Systems Inc. website: <https://baysaver.com/products/barracuda/>

**CERTIFIED FULL CAPTURE SYSTEM LIST OF  
TRASH TREATMENT CONTROL DEVICES**

| <b>Owner</b>                                      | <b>Full Capture System Trash<br/>Device Brand Name</b>                    | <b>Date Application<br/>Certified or Fact<br/>Sheet Updated</b> | <b>Date Vector<br/>Control<br/>Accessibility<br/>Verified</b> |
|---------------------------------------------------|---------------------------------------------------------------------------|-----------------------------------------------------------------|---------------------------------------------------------------|
| <b>Coanda Inc.</b>                                | <b>Coanda Trash Screen and<br/>Debris Fence</b>                           | <b>COA-1<br/>Updated 09/10/21</b>                               | <b>09/07/21</b>                                               |
| <b>Contech Construction Products<sup>22</sup></b> | <b>Continuous Deflective<br/>Separator Hydrodynamic<br/>Separator</b>     | <b>CCP-1HF<br/>Updated 05/27/21</b>                             | <b>04/29/21</b>                                               |
| <b>Jensen Stormwater Systems<sup>23</sup></b>     | <b>Jensen® Deflective<br/>Separators</b>                                  | <b>Application 5<br/>03/15/18</b>                               | <b>12/06/19</b>                                               |
| <b>Hydro International<sup>24</sup></b>           | <b>Downstream Defender (In-<br/>Line and Off-Line<br/>Configurations)</b> | <b>Application 14<br/>07/10/18</b>                              | <b>03/16/20</b>                                               |
| <b>Hydro International</b>                        | <b>First Defense® High-<br/>Capacity Full Trash Capture<br/>Device</b>    | <b>Application 28<br/>10/30/20</b>                              | <b>08/20/20</b>                                               |

<sup>22</sup> Contech Construction Products website: <http://www.conteches.com/products/stormwater-management/treatment/cds/>

<sup>23</sup> Jensen Stormwater Systems website: <http://www.jensenengineeredsystems.com/>

<sup>24</sup> Hydro International website: <https://www.hydro-int.com/>



**CERTIFIED FULL CAPTURE SYSTEM LIST OF  
TRASH TREATMENT CONTROL DEVICES**

| <b>Owner</b>                            | <b>Full Capture System Trash<br/>Device Brand Name</b>                             | <b>Date Application<br/>Certified or Fact<br/>Sheet Updated</b> | <b>Date Vector<br/>Control<br/>Accessibility<br/>Verified</b> |
|-----------------------------------------|------------------------------------------------------------------------------------|-----------------------------------------------------------------|---------------------------------------------------------------|
| <b>Hydro International</b>              | <b>Hydro Up-Flo Filter®</b>                                                        | <b>Application 11<br/>07/18/18</b>                              | <b>03/16/20</b>                                               |
| <b>Hydro International</b>              | <b>Hydro DryScreen</b>                                                             | <b>Application 10<br/>07/10/18<br/>Updated 05/05/21</b>         | <b>04/29/21</b>                                               |
| <b>Oldcastle Infrastructure</b>         | <b>FloGard® NetTech</b>                                                            | <b>OI-11HF<br/>Updated 12/08/20</b>                             | <b>12/03/20</b>                                               |
| <b>Oldcastle Infrastructure</b>         | <b>Nutrient Separating Baffle<br/>Box®</b>                                         | <b>Application 17<br/>10/12/18<br/>Updated 07/21/20</b>         | <b>05/01/20</b>                                               |
| <b>Roscoe Moss Company<sup>25</sup></b> | <b>Storm Flo® Trash Screen –<br/>Linear Radial Gross Solids<br/>Removal Device</b> | <b>RMC-1HF<br/>Updated 03/30/21</b>                             | <b>03/11/21</b>                                               |

<sup>25</sup> Roscoe Moss Company website: <https://roscoemoss.com/products/stormwater-gross-solids-removal-device/>

**CERTIFIED FULL CAPTURE SYSTEM LIST OF  
TRASH TREATMENT CONTROL DEVICES**

| <b>Owner</b>                                                              | <b>Full Capture System Trash<br/>Device Brand Name</b> | <b>Date Application<br/>Certified or Fact<br/>Sheet Updated</b> | <b>Date Vector<br/>Control<br/>Accessibility<br/>Verified</b> |
|---------------------------------------------------------------------------|--------------------------------------------------------|-----------------------------------------------------------------|---------------------------------------------------------------|
| <b>StormTrap Modular Concrete Stormwater<br/>Management</b> <sup>26</sup> | <b>SiteSaver®</b>                                      | <b>Application 9<br/>03/15/18<br/>Updated 02/23/21</b>          | <b>03/18/21</b>                                               |

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<sup>26</sup> StormTrap <https://stormtrap.com>



## **ATTACHMENT 3**

Detailed Costs and Calculations

*(provided electronically)*

## **ATTACHMENT 4**

Permitting Matrix

| Site Name | Biological Resources                                                                                                                                                                                                                                                    | Cultural Resources                                                                                                              | Geology & Soils                                                                                                                                                                                                                                                                                                                            | Hydrology & Water Quality                                                                                                                                                                                                                                                                                                                              | Transportation                                                                                                                                                                                                                                                    | Tribal Cultural Resources                                                                                                                                                                                                                                                                                                                     | Utilities                                                                                                                                                                                         | CEQA Cost    | Biological Technical Memo/Report Cost | Cultural Resources Inventory Report Cost | Aquatic Resources Delineation Cost | USACE Section 404 Permit Cost | RWQCB Section 401 Permit Cost | CDFW LSAA Permit Cost | BCDC Permit Cost | Total Permitting / CEQA Cost | Notes                                                                                                                                                                                         |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---------------------------------------|------------------------------------------|------------------------------------|-------------------------------|-------------------------------|-----------------------|------------------|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CM-016    | No special status species or habitats were identified within the project site. Sensitive habitats were identified in the project buffer area. A biological technical memo should be sufficient for CEQA purposes.                                                       | The California Historical Resources Information System records search for the project area was negative for cultural resources. | This site lies on soil identified as bay mud below a shallow layer of fill with a high groundwater table. This alternative would require excavations of up to 25 feet and potentially significant dewatering during construction. Due to the presence of bay mud this alternative may require over excavation or the possible use of fill. | The site lies within a special flood hazard Zone AE. The area contains a high groundwater level; therefore, extensive dewatering could be required during construction. No groundwater contamination is known to exist in the area per Geotracker. If concrete is installed, pouring and curing would have to follow the State Water Board procedures. | During construction, roadways could be closed and detours implemented for approximately 2-4 weeks during construction. Once constructed, maintenance on the trash capture device could cause traffic flow issues for approximately 2-4 hours, 2-3 times per year. | The NAHC database search response was received on March 11, 2022, and indicated that there are Native American Resources in the vicinity of the sites. The response, however, did not indicate where the site(s) are located. Further correspondence with the Graton Rancheria will be required to determine if this site is near a resource. | This alternative could affect existing underground utilities located within the project sites. Utility relocation would require greater excavation and increase tribal cultural resource impacts. | \$ 5,000.00  | \$ 8,000.00                           | \$ 14,000.00                             | \$ -                               | \$ -                          | \$ -                          | \$ -                  | \$ -             | \$ 27,000.00                 | Underground in roadway. Depending on results of cultural survey (if no issues identified), cultural resources inventory report fee may be reduced significantly.                              |
| LS-010    | No special status species or habitats were identified within the project site. Sensitive habitats were identified in the project buffer area. A biological technical memo should be sufficient for CEQA purposes.                                                       | The California Historical Resources Information System records search for the project area was negative for cultural resources. | This site lies on soil identified as bay mud below a shallow layer of fill with a high groundwater table. This alternative would require excavations of up to 25 feet and potentially significant dewatering during construction. Due to the presence of bay mud this alternative may require over excavation or the possible use of fill. | The site lies within a special flood hazard Zone AE. The area contains a high groundwater level; therefore, extensive dewatering could be required during construction. No groundwater contamination is known to exist in the area per Geotracker. If concrete is installed, pouring and curing would have to follow the State Water Board procedures. | During construction, roadways could be closed and detours implemented for approximately 2-4 weeks during construction. Once constructed, maintenance on the trash capture device could cause traffic flow issues for approximately 2-4 hours, 2-3 times per year. | The NAHC database search response was received on March 11, 2022, and indicated that there are Native American Resources in the vicinity of the sites. The response, however, did not indicate where the site(s) are located. Further correspondence with the Graton Rancheria will be required to determine if this site is near a resource. | This alternative could affect existing underground utilities located within the project sites. Utility relocation would require greater excavation and increase tribal cultural resource impacts. | \$ 5,000.00  | \$ 8,000.00                           | \$ 14,000.00                             | \$ -                               | \$ -                          | \$ -                          | \$ -                  | \$ -             | \$ 27,000.00                 | Underground in roadway. Depending on results of cultural survey (if no issues identified), cultural resources inventory report fee may be reduced significantly.                              |
| LS-014    | No special status species or habitats were identified within the project site. Sensitive habitats were identified in the project buffer area. A biological technical memo should be sufficient for CEQA purposes.                                                       | The California Historical Resources Information System records search for the project area was negative for cultural resources. | This site lies on soil identified as bay mud below a shallow layer of fill with a high groundwater table. This alternative would require excavations of up to 25 feet and potentially significant dewatering during construction. Due to the presence of bay mud this alternative may require over excavation or the possible use of fill. | The site lies within a special flood hazard Zone AE. The area contains a high groundwater level; therefore, extensive dewatering could be required during construction. No groundwater contamination is known to exist in the area per Geotracker. If concrete is installed, pouring and curing would have to follow the State Water Board procedures. | During construction, roadways could be closed and detours implemented for approximately 2-4 weeks during construction. Once constructed, maintenance on the trash capture device could cause traffic flow issues for approximately 2-4 hours, 2-3 times per year. | The NAHC database search response was received on March 11, 2022, and indicated that there are Native American Resources in the vicinity of the sites. The response, however, did not indicate where the site(s) are located. Further correspondence with the Graton Rancheria will be required to determine if this site is near a resource. | This alternative could affect existing underground utilities located within the project sites. Utility relocation would require greater excavation and increase tribal cultural resource impacts. | \$ 5,000.00  | \$ 8,000.00                           | \$ 14,000.00                             | \$ -                               | \$ -                          | \$ -                          | \$ -                  | \$ -             | \$ 27,000.00                 | Underground in roadway. Depending on results of cultural survey (if no issues identified), cultural resources inventory report fee may be reduced significantly.                              |
| LS-028    | No special status species were identified at this site. However, special status species may be present, and project activities would occur with protected aquatic resources. Therefore, a full biological resources assessment report would be required for permitting. | The California Historical Resources Information System records search for the project area was negative for cultural resources. | This site lies on soil identified as bay mud below a shallow layer of fill with a high groundwater table. This alternative would require excavations of up to 25 feet and potentially significant dewatering during construction. Due to the presence of bay mud this alternative may require over excavation or the possible use of fill. | The site lies within a special flood hazard Zone AE. The area contains a high groundwater level; therefore, extensive dewatering could be required during construction. No groundwater contamination is known to exist in the area per Geotracker. If concrete is installed, pouring and curing would have to follow the State Water Board procedures. | During construction, roadways could be closed and detours implemented for approximately 2-4 weeks during construction. Once constructed, maintenance on the trash capture device could cause traffic flow issues for approximately 2-4 hours, 2-3 times per year. | The NAHC database search response was received on March 11, 2022, and indicated that there are Native American Resources in the vicinity of the sites. The response, however, did not indicate where the site(s) are located. Further correspondence with the Graton Rancheria will be required to determine if this site is near a resource. | This alternative could affect existing underground utilities located within the project site. Utility relocation would require greater excavation and increase tribal cultural resource impacts.  | \$ 28,000.00 | \$ 12,000.00                          | \$ 14,000.00                             | \$ 12,000.00                       | \$ 12,000.00                  | \$ 15,000.00                  | \$ 12,000.00          | \$ 15,000.00     | \$ 120,000.00                | Work in waters-tidal.                                                                                                                                                                         |
| MV-037    | No special status species or habitats were identified at this site. A biological technical memo should be sufficient for permitting.                                                                                                                                    | The California Historical Resources Information System records search for the project area was negative for cultural resources. | This site lies on soil identified as bay mud below a shallow layer of fill with a high groundwater table. This alternative would require excavations of up to 25 feet and potentially significant dewatering during construction. Due to the presence of bay mud this alternative may require over excavation or the possible use of fill. | The site lies within a special flood hazard Zone AE. The area contains a high groundwater level; therefore, extensive dewatering could be required during construction. No groundwater contamination is known to exist in the area per Geotracker. If concrete is installed, pouring and curing would have to follow the State Water Board procedures. | During construction, roadways could be closed and detours implemented for approximately 2-4 weeks during construction. Once constructed, maintenance on the trash capture device could cause traffic flow issues for approximately 2-4 hours, 2-3 times per year. | The NAHC database search response was received on March 11, 2022, and indicated that there are Native American Resources in the vicinity of the sites. The response, however, did not indicate where the site(s) are located. Further correspondence with the Graton Rancheria will be required to determine if this site is near a resource. | This alternative could affect existing underground utilities located within the project site. Utility relocation would require greater excavation and increase tribal cultural resource impacts.  | \$ 5,000.00  | \$ 8,000.00                           | \$ 14,000.00                             | \$ -                               | \$ -                          | \$ -                          | \$ -                  | \$ -             | \$ 27,000.00                 | Cement lined ditch. Depending on results of cultural survey (if no issues identified), cultural resources inventory report fee may be reduced significantly.                                  |
| NV-029    | No special status species or habitats were identified at this site. A biological technical memo should be sufficient for permitting.                                                                                                                                    | The California Historical Resources Information System records search for the project area was negative for cultural resources. | This site lies on soil identified as bay mud below a shallow layer of fill with a high groundwater table. This alternative would require excavations of up to 25 feet and potentially significant dewatering during construction. Due to the presence of bay mud this alternative may require over excavation or the possible use of fill. | The site lies within a special flood hazard Zone AE. The area contains a high groundwater level; therefore, extensive dewatering could be required during construction. No groundwater contamination is known to exist in the area per Geotracker. If concrete is installed, pouring and curing would have to follow the State Water Board procedures. | During construction, roadways could be closed and detours implemented for approximately 2-4 weeks during construction. Once constructed, maintenance on the trash capture device could cause traffic flow issues for approximately 2-4 hours, 2-3 times per year. | The NAHC database search response was received on March 11, 2022, and indicated that there are Native American Resources in the vicinity of the sites. The response, however, did not indicate where the site(s) are located. Further correspondence with the Graton Rancheria will be required to determine if this site is near a resource. | This alternative could affect existing underground utilities located within the project sites. Utility relocation would require greater excavation and increase tribal cultural resource impacts. | \$ 5,000.00  | \$ 8,000.00                           | \$ 14,000.00                             | \$ -                               | \$ -                          | \$ -                          | \$ -                  | \$ -             | \$ 27,000.00                 | Storm drain only, CEQA Categorical Exclusion likely. Depending on results of cultural survey (if no issues identified), cultural resources inventory report fee may be reduced significantly. |



| Site Name | Biological Resources                                                                                                                                                                                                                                                    | Cultural Resources                                                                                                              | Geology & Soils                                                                                                                                                                                                                                                                                                                            | Hydrology & Water Quality                                                                                                                                                                                                                                                                                                                              | Transportation                                                                                                                                                                                                                                                    | Tribal Cultural Resources                                                                                                                                                                                                                                                                                                                     | Utilities                                                                                                                                                                                        | CEQA Cost    | Biological Technical Memo/Report Cost | Cultural Resources Inventory Report Cost | Aquatic Resources Delineation Cost | USACE Section 404 Permit Cost | RWQCB Section 401 Permit Cost | CDFW LSAA Permit Cost | BCDC Permit Cost | Total Permitting / CEQA Cost | Notes                                                                                                                                                                                |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---------------------------------------|------------------------------------------|------------------------------------|-------------------------------|-------------------------------|-----------------------|------------------|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NV-116    | No special status species were identified at this site. However, special status species may be present, and project activities would occur with protected aquatic resources. Therefore, a full biological resources assessment report would be required for permitting. | The California Historical Resources Information System records search for the project area was negative for cultural resources. | This site lies on soil identified as bay mud below a shallow layer of fill with a high groundwater table. This alternative would require excavations of up to 25 feet and potentially significant dewatering during construction. Due to the presence of bay mud this alternative may require over excavation or the possible use of fill. | The site lies within a special flood hazard Zone AE. The area contains a high groundwater level; therefore, extensive dewatering could be required during construction. No groundwater contamination is known to exist in the area per Geotracker. If concrete is installed, pouring and curing would have to follow the State Water Board procedures. | During construction, roadways could be closed and detours implemented for approximately 2-4 weeks during construction. Once constructed, maintenance on the trash capture device could cause traffic flow issues for approximately 2-4 hours, 2-3 times per year. | The NAHC database search response was received on March 11, 2022, and indicated that there are Native American Resources in the vicinity of the sites. The response, however, did not indicate where the site(s) are located. Further correspondence with the Graton Rancheria will be required to determine if this site is near a resource. | This alternative could affect existing underground utilities located within the project site. Utility relocation would require greater excavation and increase tribal cultural resource impacts. | \$ 28,000.00 | \$ 12,000.00                          | \$ 14,000.00                             | \$ 12,000.00                       | \$ 12,000.00                  | \$ 15,000.00                  | \$ 12,000.00          | \$ -             | \$ 105,000.00                | Non-tidal ditch.                                                                                                                                                                     |
| SR-108    | No special status species or habitats were identified at this site. A biological technical memo should be sufficient for permitting.                                                                                                                                    | The California Historical Resources Information System records search for the project area was negative for cultural resources. | This site lies on soil identified as bay mud below a shallow layer of fill with a high groundwater table. This alternative would require excavations of up to 25 feet and potentially significant dewatering during construction. Due to the presence of bay mud this alternative may require over excavation or the possible use of fill. | The site lies within a special flood hazard Zone AE. The area contains a high groundwater level; therefore, extensive dewatering could be required during construction. No groundwater contamination is known to exist in the area per Geotracker. If concrete is installed, pouring and curing would have to follow the State Water Board procedures. | During construction, roadways could be closed and detours implemented for approximately 2-4 weeks during construction. Once constructed, maintenance on the trash capture device could cause traffic flow issues for approximately 2-4 hours, 2-3 times per year. | The NAHC database search response was received on March 11, 2022, and indicated that there are Native American Resources in the vicinity of the sites. The response, however, did not indicate where the site(s) are located. Further correspondence with the Graton Rancheria will be required to determine if this site is near a resource. | This alternative could affect existing underground utilities located within the project site. Utility relocation would require greater excavation and increase tribal cultural resource impacts. | \$ 5,000.00  | \$ 8,000.00                           | \$ 14,000.00                             | \$ -                               | \$ -                          | \$ -                          | \$ -                  | \$ -             | \$ 27,000.00                 | In upland area outside of San Rafael Creek. Depending on results of cultural survey (if no issues identified), cultural resources inventory report fee may be reduced significantly. |
| SR-111    | No special status species were identified at this site. However, special status species may be present, and project activities would occur with protected aquatic resources. Therefore, a full biological resources assessment report would be required for permitting. | The California Historical Resources Information System records search for the project area was negative for cultural resources. | This site lies on soil identified as bay mud below a shallow layer of fill with a high groundwater table. This alternative would require excavations of up to 25 feet and potentially significant dewatering during construction. Due to the presence of bay mud this alternative may require over excavation or the possible use of fill. | The site lies within a special flood hazard Zone AE. The area contains a high groundwater level; therefore, extensive dewatering could be required during construction. No groundwater contamination is known to exist in the area per Geotracker. If concrete is installed, pouring and curing would have to follow the State Water Board procedures. | During construction, roadways could be closed and detours implemented for approximately 2-4 weeks during construction. Once constructed, maintenance on the trash capture device could cause traffic flow issues for approximately 2-4 hours, 2-3 times per year. | The NAHC database search response was received on March 11, 2022, and indicated that there are Native American Resources in the vicinity of the sites. The response, however, did not indicate where the site(s) are located. Further correspondence with the Graton Rancheria will be required to determine if this site is near a resource. | This alternative could affect existing underground utilities located within the project site. Utility relocation would require greater excavation and increase tribal cultural resource impacts. | \$ 28,000.00 | \$ 8,000.00                           | \$ 14,000.00                             | \$ 12,000.00                       | \$ 12,000.00                  | \$ 15,000.00                  | \$ 12,000.00          | \$ 15,000.00     | \$ 116,000.00                | Work in waters-tidal.                                                                                                                                                                |
| SR-129    | No special status species were identified at this site. However, special status species may be present, and project activities would occur with protected aquatic resources. Therefore, a full biological resources assessment report would be required for permitting. | The California Historical Resources Information System records search for the project area was negative for cultural resources. | This site lies on soil identified as bay mud below a shallow layer of fill with a high groundwater table. This alternative would require excavations of up to 25 feet and potentially significant dewatering during construction. Due to the presence of bay mud this alternative may require over excavation or the possible use of fill. | The site lies within a special flood hazard Zone AE. The area contains a high groundwater level; therefore, extensive dewatering could be required during construction. No groundwater contamination is known to exist in the area per Geotracker. If concrete is installed, pouring and curing would have to follow the State Water Board procedures. | During construction, roadways could be closed and detours implemented for approximately 2-4 weeks during construction. Once constructed, maintenance on the trash capture device could cause traffic flow issues for approximately 2-4 hours, 2-3 times per year. | The NAHC database search response was received on March 11, 2022, and indicated that there are Native American Resources in the vicinity of the sites. The response, however, did not indicate where the site(s) are located. Further correspondence with the Graton Rancheria will be required to determine if this site is near a resource. | This alternative could affect existing underground utilities located within the project site. Utility relocation would require greater excavation and increase tribal cultural resource impacts. | \$ 28,000.00 | \$ 12,000.00                          | \$ 14,000.00                             | \$ 12,000.00                       | \$ 12,000.00                  | \$ 15,000.00                  | \$ 12,000.00          | \$ -             | \$ 105,000.00                | Non tidal aquatic habitat in project site.                                                                                                                                           |
| SS-002    | No special status species or habitats were identified at this site. A biological technical memo should be sufficient for permitting.                                                                                                                                    | The California Historical Resources Information System records search for the project area was negative for cultural resources. | This site lies on soil identified as bay mud below a shallow layer of fill with a high groundwater table. This alternative would require excavations of up to 25 feet and potentially significant dewatering during construction. Due to the presence of bay mud this alternative may require over excavation or the possible use of fill. | The site lies within a special flood hazard Zone AE. The area contains a high groundwater level; therefore, extensive dewatering could be required during construction. No groundwater contamination is known to exist in the area per Geotracker. If concrete is installed, pouring and curing would have to follow the State Water Board procedures. | During construction, roadways could be closed and detours implemented for approximately 2-4 weeks during construction. Once constructed, maintenance on the trash capture device could cause traffic flow issues for approximately 2-4 hours, 2-3 times per year. | The NAHC database search response was received on March 11, 2022, and indicated that there are Native American Resources in the vicinity of the sites. The response, however, did not indicate where the site(s) are located. Further correspondence with the Graton Rancheria will be required to determine if this site is near a resource. | This alternative could affect existing underground utilities located within the project site. Utility relocation would require greater excavation and increase tribal cultural resource impacts. | \$ 5,000.00  | \$ 8,000.00                           | \$ 14,000.00                             | \$ -                               | \$ -                          | \$ -                          | \$ -                  | \$ -             | \$ 27,000.00                 | CEQA Categorical Exclusion likely. Depending on results of cultural survey (if no issues identified), cultural resources inventory report fee may be reduced significantly.          |
| SS-003    | No special status species or habitats were identified at this site. A biological technical memo should be sufficient for permitting.                                                                                                                                    | The California Historical Resources Information System records search for the project area was negative for cultural resources. | This site lies on soil identified as bay mud below a shallow layer of fill with a high groundwater table. This alternative would require excavations of up to 25 feet and potentially significant dewatering during construction. Due to the presence of bay mud this alternative may require over excavation or the possible use of fill. | The site lies within a special flood hazard Zone AE. The area contains a high groundwater level; therefore, extensive dewatering could be required during construction. No groundwater contamination is known to exist in the area per Geotracker. If concrete is installed, pouring and curing would have to follow the State Water Board procedures. | During construction, roadways could be closed and detours implemented for approximately 2-4 weeks during construction. Once constructed, maintenance on the trash capture device could cause traffic flow issues for approximately 2-4 hours, 2-3 times per year. | The NAHC database search response was received on March 11, 2022, and indicated that there are Native American Resources in the vicinity of the sites. The response, however, did not indicate where the site(s) are located. Further correspondence with the Graton Rancheria will be required to determine if this site is near a resource. | This alternative could affect existing underground utilities located within the project site. Utility relocation would require greater excavation and increase tribal cultural resource impacts. | \$ 5,000.00  | \$ 8,000.00                           | \$ 14,000.00                             | \$ -                               | \$ -                          | \$ -                          | \$ -                  | \$ -             | \$ 27,000.00                 | CEQA Categorical Exclusion likely. Depending on results of cultural survey (if no issues identified), cultural resources inventory report fee may be reduced significantly.          |

| Site Name | Biological Resources                                                                                                                                                                                                                                                    | Cultural Resources                                                                                                              | Geology & Soils                                                                                                                                                                                                                                                                                                                            | Hydrology & Water Quality                                                                                                                                                                                                                                                                                                                              | Transportation                                                                                                                                                                                                                                                    | Tribal Cultural Resources                                                                                                                                                                                                                                                                                                                     | Utilities                                                                                                                                                                                        | CEQA Cost    | Biological Technical Memo/Report Cost | Cultural Resources Inventory Report Cost | Aquatic Resources Delineation Cost | USACE Section 404 Permit Cost | RWQCB Section 401 Permit Cost | CDFW LSAA Permit Cost | BCDC Permit Cost | Total Permitting / CEQA Cost | Notes                                                                                                                                                                                                                                                     |
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| ST-003    | No special status species were identified at this site. However, special status species may be present, and project activities would occur with protected aquatic resources. Therefore, a full biological resources assessment report would be required for permitting. | The California Historical Resources Information System records search for the project area was negative for cultural resources. | This site lies on soil identified as bay mud below a shallow layer of fill with a high groundwater table. This alternative would require excavations of up to 25 feet and potentially significant dewatering during construction. Due to the presence of bay mud this alternative may require over excavation or the possible use of fill. | The site lies within a special flood hazard Zone AE. The area contains a high groundwater level; therefore, extensive dewatering could be required during construction. No groundwater contamination is known to exist in the area per Geotracker. If concrete is installed, pouring and curing would have to follow the State Water Board procedures. | This alternative would not have an significant effect on transportation.                                                                                                                                                                                          | The NAHC database search response was received on March 11, 2022, and indicated that there are Native American Resources in the vicinity of the sites. The response, however, did not indicate where the site(s) are located. Further correspondence with the Graton Rancheria will be required to determine if this site is near a resource. | This alternative could affect existing underground utilities located within the project site. Utility relocation would require greater excavation and increase tribal cultural resource impacts. | \$ 28,000.00 | \$ 12,000.00                          | \$ 14,000.00                             | \$ 12,000.00                       | \$ 12,000.00                  | \$ 15,000.00                  | \$ 12,000.00          | \$ 15,000.00     | \$ 120,000.00                | Work in waters-tidal.                                                                                                                                                                                                                                     |
| TB-010    | No special status species or habitats were identified at this site. A biological technical memo should be sufficient for permitting.                                                                                                                                    | The California Historical Resources Information System records search for the project area was negative for cultural resources. | This site lies on soil identified as bay mud below a shallow layer of fill with a high groundwater table. This alternative would require excavations of up to 25 feet and potentially significant dewatering during construction. Due to the presence of bay mud this alternative may require over excavation or the possible use of fill. | The site lies within a special flood hazard Zone AE. The area contains a high groundwater level; therefore, extensive dewatering could be required during construction. No groundwater contamination is known to exist in the area per Geotracker. If concrete is installed, pouring and curing would have to follow the State Water Board procedures. | During construction, roadways could be closed and detours implemented for approximately 2-4 weeks during construction. Once constructed, maintenance on the trash capture device could cause traffic flow issues for approximately 2-4 hours, 2-3 times per year. | The NAHC database search response was received on March 11, 2022, and indicated that there are Native American Resources in the vicinity of the sites. The response, however, did not indicate where the site(s) are located. Further correspondence with the Graton Rancheria will be required to determine if this site is near a resource. | This alternative could affect existing underground utilities located within the project site. Utility relocation would require greater excavation and increase tribal cultural resource impacts. | \$ 5,000.00  | \$ 8,000.00                           | \$ 14,000.00                             | \$ -                               | \$ -                          | \$ -                          | \$ -                  | \$ -             | \$ 27,000.00                 | CEQA Categorical Exclusion likely. Depending on results of cultural survey (if no issues identified), cultural resources inventory report fee may be reduced significantly.                                                                               |
| TV-001    | No special status species or habitats were identified at this site. A biological technical memo should be sufficient for permitting.                                                                                                                                    | The California Historical Resources Information System records search for the project area was negative for cultural resources. | This site lies on soil identified as bay mud below a shallow layer of fill with a high groundwater table. This alternative would require excavations of up to 25 feet and potentially significant dewatering during construction. Due to the presence of bay mud this alternative may require over excavation or the possible use of fill. | The site lies within a special flood hazard Zone AE. The area contains a high groundwater level; therefore, extensive dewatering could be required during construction. No groundwater contamination is known to exist in the area per Geotracker. If concrete is installed, pouring and curing would have to follow the State Water Board procedures. | This alternative is located in a existing pump station and would not have an significant effect on transportation.                                                                                                                                                | The NAHC database search response was received on March 11, 2022, and indicated that there are Native American Resources in the vicinity of the sites. The response, however, did not indicate where the site(s) are located. Further correspondence with the Graton Rancheria will be required to determine if this site is near a resource. | This alternative could affect existing underground utilities located within the project site. Utility relocation would require greater excavation and increase tribal cultural resource impacts. | \$ 5,000.00  | \$ 8,000.00                           | \$ 14,000.00                             | \$ -                               | \$ -                          | \$ -                          | \$ -                  | \$ -             | \$ 27,000.00                 | Located in pump station or bike trail. Not located within wetlands or stream. CEQA Categorical Exclusion likely. Depending on results of cultural survey (if no issues identified), cultural resources inventory report fee may be reduced significantly. |
| SR-143    | No special status species or habitats were identified at this site. A biological technical memo should be sufficient for permitting.                                                                                                                                    | The California Historical Resources Information System records search for the project area was negative for cultural resources. | This site lies on soil identified as bay mud below a shallow layer of fill with a high groundwater table. This alternative would require excavations of up to 25 feet and potentially significant dewatering during construction. Due to the presence of bay mud this alternative may require over excavation or the possible use of fill. | The site lies within a special flood hazard Zone AE. The area contains a high groundwater level; therefore, extensive dewatering could be required during construction. No groundwater contamination is known to exist in the area per Geotracker. If concrete is installed, pouring and curing would have to follow the State Water Board procedures. | This alternative is located in a existing pump station and would not have an significant effect on transportation.                                                                                                                                                | The NAHC database search response was received on March 11, 2022, and indicated that there are Native American Resources in the vicinity of the sites. The response, however, did not indicate where the site(s) are located. Further correspondence with the Graton Rancheria will be required to determine if this site is near a resource. | This alternative could affect existing underground utilities located within the project site. Utility relocation would require greater excavation and increase tribal cultural resource impacts. | \$ 5,000.00  | \$ 8,000.00                           | \$ 14,000.00                             | \$ -                               | \$ -                          | \$ -                          | \$ -                  | \$ -             | \$ 27,000.00                 | At pump station but screens will be cantilevered over water so no in water work. Depending on results of cultural survey (if no issues identified), cultural resources inventory report fee may be reduced significantly.                                 |

**General Assumptions:**  
1. CEQA may require MND (unless noted otherwise). Depends on issues that come up. Can group multiple sites as one project to significantly reduce costs.  
2. Technical reporting and permitting costs can also be reduced significantly if multiple sites are Included within a single project.  
3. No Coastal Development Permits required as none of the sites fall under CCC jurisdiction.  
4. BCDC permitting required when site is within 100 foot band of San Francisco Bay.

## **ATTACHMENT 5**

GIS Shapefile for Small Device Installation

*(provided electronically)*